

Mathematics Stage 5 Path

Non-Right-Angled Trigonometry

Book 2 Apply the sine and cosine rules to any triangle and solve related problems

Version: 260905

Feedback:
<https://MrDingMaths.com>

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Syllabus Content

MA5-TRG-P-01 applies Pythagoras' theorem and trigonometry to solve 3-dimensional problems and applies the sine, cosine and area rules to solve 2-dimensional problems, including bearings

Apply the sine, cosine and area rules to any triangle and solve related problems

- Use graphing applications to verify the sine rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ and that the ratios of a side to the sine of the opposite angle is a constant
- Apply the sine rule in a given triangle ABC to find the value of an unknown side
- Apply the sine rule in a given triangle ABC to find the value of an unknown angle (ambiguous case excluded): $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$
- Use graphing applications to verify the cosine rule $c^2 = a^2 + b^2 - 2ab \cos C$.
- Apply the cosine rule to find the unknown sides for a given triangle ABC .
- Rearrange the formula to deduce that $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$ and use this to find an unknown angle
- Use graphing applications to verify the area rule $A = \frac{1}{2}ab \sin C$.
- Apply the formula $A = \frac{1}{2}ab \sin C$, where a and b are the sides that form angle C to find the area of a given triangle ABC
- Solve problems involving finding unknown angles or sides in triangles (excluding right-angled triangles) by selecting and applying the appropriate rule

Labelling Triangles

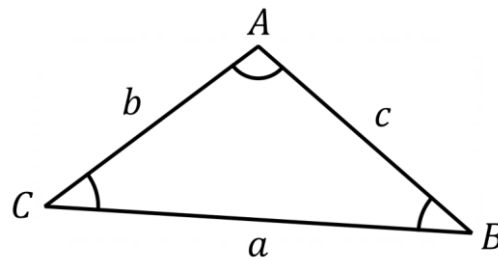
i Labelling Triangles

In a triangle, lowercase letters represent sides and uppercase letters represent angles.

Each side is named with the same letter as the angle **opposite** to it.

Side a is opposite angle A . Side b is opposite angle B . Side c is opposite angle C .

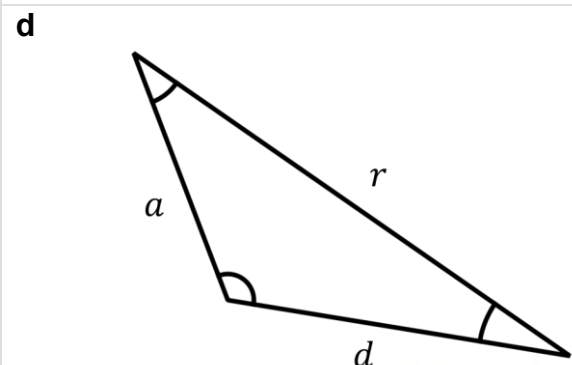
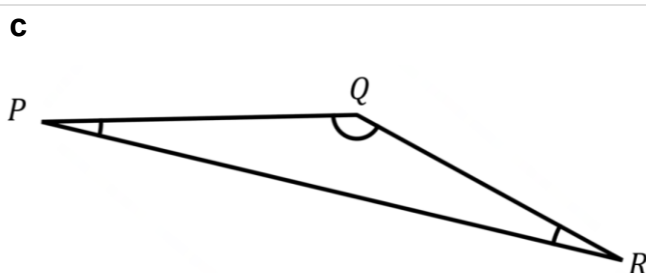
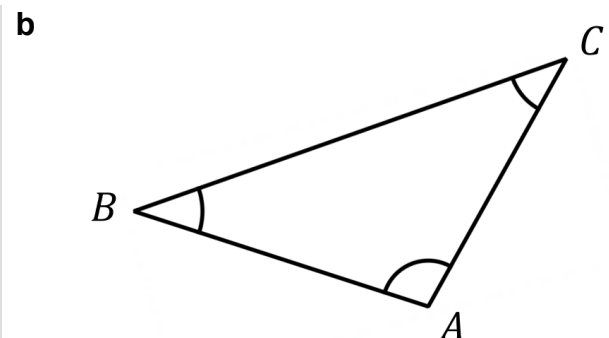
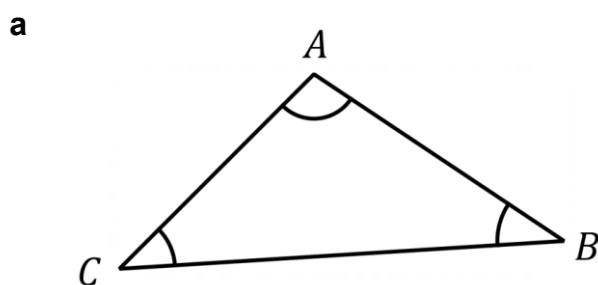
Opposite sides and angles do not touch.



⚠ We label a vertex (corner) of a triangle and the angle at that vertex using the same uppercase letter.

? Identify Triangle opposite angle/side pairs

Label the missing sides/angles of these triangles.

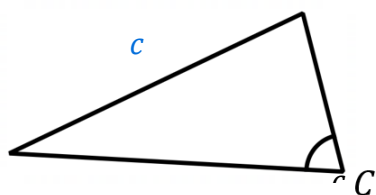


💡 Key Ideas

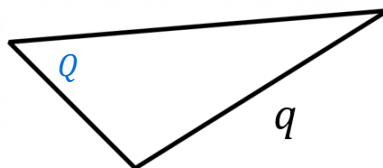
1. In a triangle, each side is named with the same letter as its opposite angle.
2. Side a is opposite angle, side is opposite angle B .
3. To find which angle is opposite a side, look for the angle that the side does NOT

1

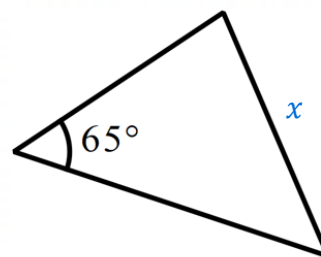
a Label the side opposite angle C .



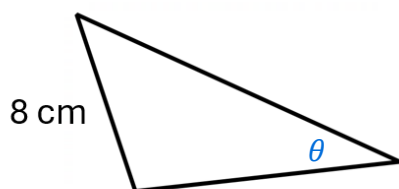
b Label the angle opposite side q



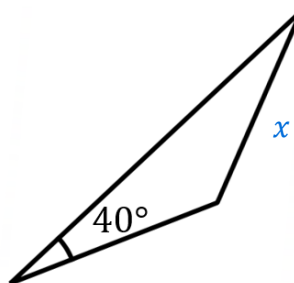
c Label the side opposite the 65° angle x



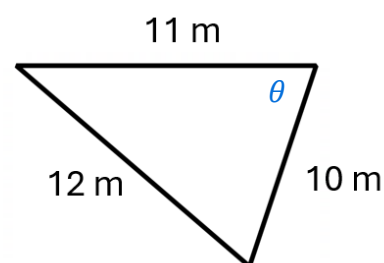
d Use θ to label the angle opposite the 8 cm side



e Use x to label the side opposite the 40° angle

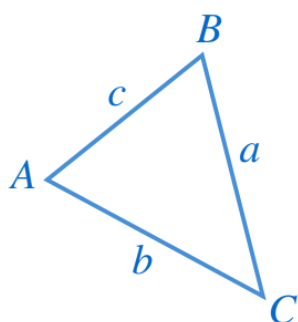


f Use θ to label the angle opposite the 12 m side

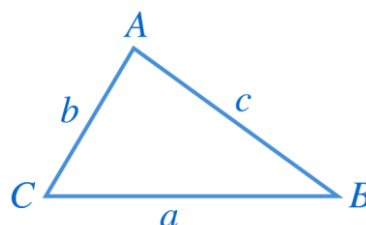


2 Label the missing sides and angles of these triangles.

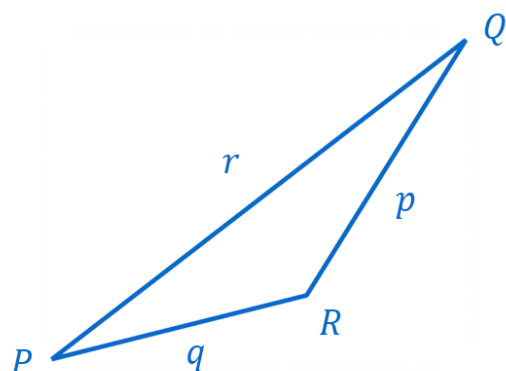
a



b



c

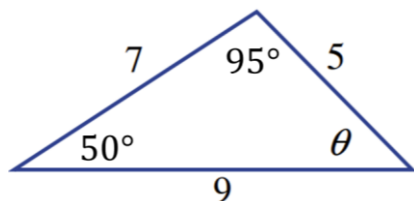


Sine Rule for Sides

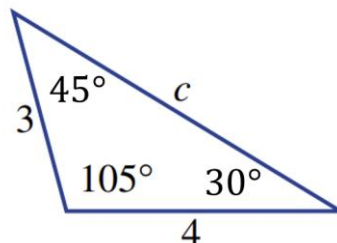
Review

- ☐ Identify the side/angle opposite in a triangle

a Which side is opposite angle θ ?



b Which angle is opposite side c ?



- ☐ Solve proportion equations
Round answers to 1 d.p. where necessary

a

$$\frac{x}{7} = \frac{12}{5}$$

b

$$\frac{8}{x} = \frac{15}{11}$$

c

$$\frac{x}{\sin 40^\circ} = \frac{8}{\sin 65^\circ}$$

d

$$\frac{x}{\sin 35^\circ} = \frac{12}{\sin 48^\circ}$$

e

$$\frac{22}{\sin 45^\circ} = \frac{x}{\sin 60^\circ}$$

f

$$\frac{\sin 72^\circ}{15} = \frac{\sin 35^\circ}{x}$$

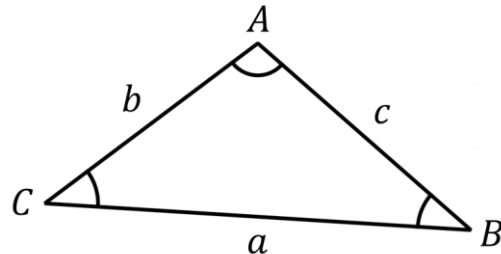
① Sine Rule

In any triangle:

- The longest side is opposite the largest angle.
- The shortest side is opposite the smallest angle.

We express this relationship mathematically using the **sine rule**:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

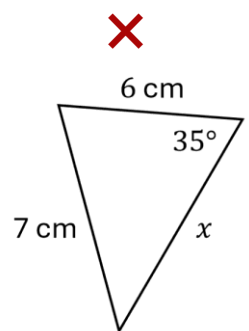
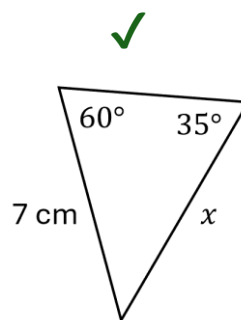
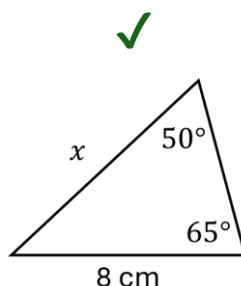
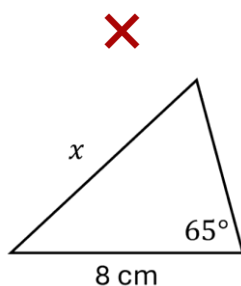


To use the sine rule to find a side, you need:

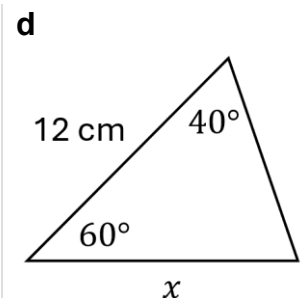
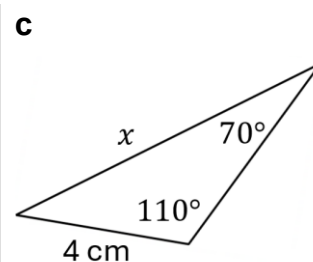
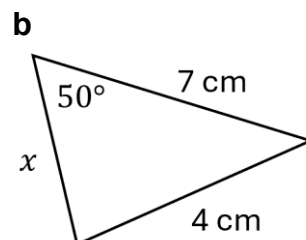
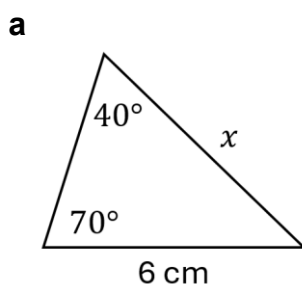
- A complete 'opposite pair' – one side AND its opposite angle (both known).
- The opposite angle to the side you want to find.

⚠ You need just TWO fractions from the sine rule: one for the known pair, one for the unknown side.

② Identify whether you can use sine rule to find a side



Identify whether you can use sine rule to find the missing side.

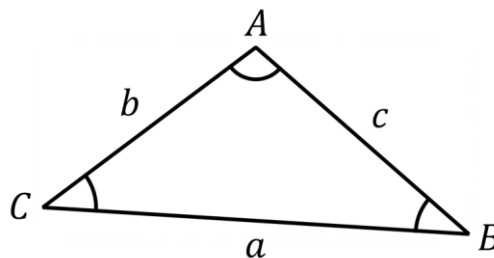


💡 Key Ideas

1. To use the sine rule to find a side, you need a complete and the opposite the side you want to find.
2. An '.....' consists of a and the opposite to it.

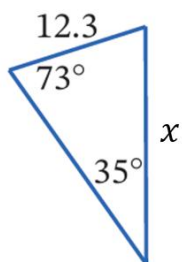
① Sine Rule (Finding a Side)

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



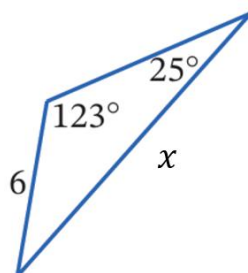
② Write sine rule equation from a diagram

a

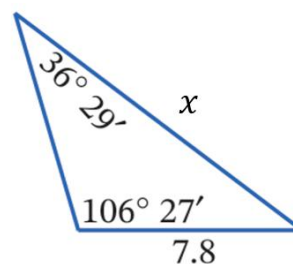


$$\frac{x}{\sin 73^\circ} = \frac{12.3}{\sin 35^\circ}$$

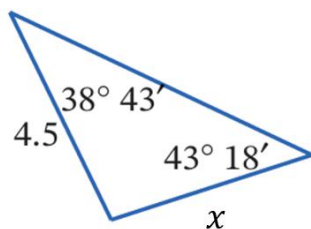
b



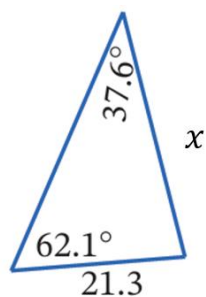
c



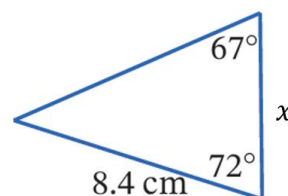
d



e

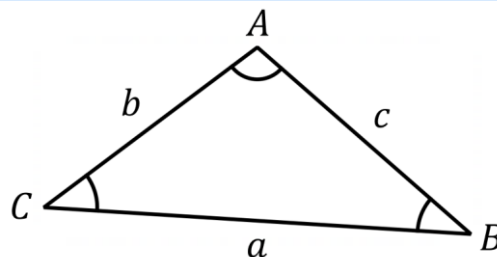


f



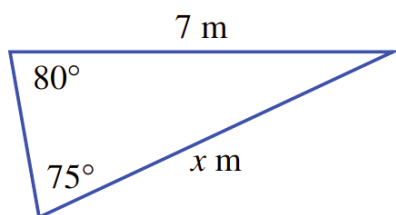
i Sine Rule (Finding a Side)

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



Example Calculate a side length using sine rule

Find the length of the missing side, to 1 d.p.

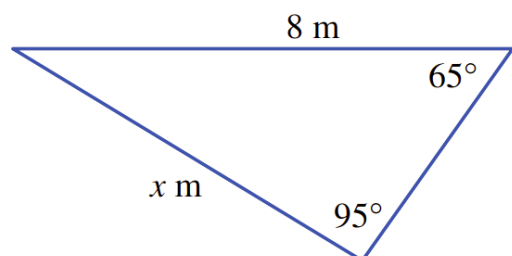


$$\begin{aligned} \frac{x}{\sin 80^\circ} &= \frac{7}{\sin 75^\circ} \\ \times \sin 80^\circ &\quad \times \sin 80^\circ \\ x &= \frac{7 \sin 80^\circ}{\sin 75^\circ} \\ &\approx 7.1 \text{ m} \end{aligned}$$

Guided Practice

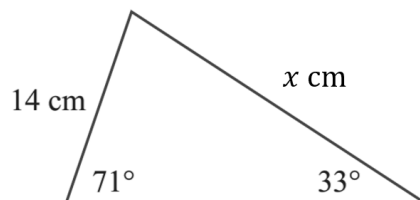
Find the length of the missing side, to 1 d.p.

a



7.3 m

b



24.3 cm

1 Solve each equation for the unknown, correct to 2 decimal places.

a

$$\frac{a}{\sin 47^\circ} = \frac{2}{\sin 51^\circ}$$

1.88

b

$$\frac{b}{\sin 31^\circ} = \frac{7}{\sin 84^\circ}$$

3.63

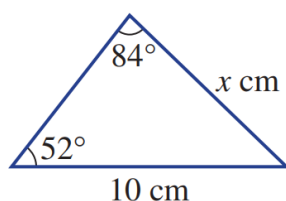
c

$$\frac{5}{\sin 63^\circ} = \frac{c}{\sin 27^\circ}$$

2.55

2 Find the value of the unknown in these triangles, correct to 1 decimal place.

a



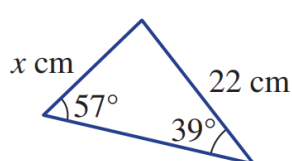
$$\frac{x}{\sin 52^\circ} = \frac{10}{\sin 84^\circ}$$

$$x = \dots\dots\dots$$

$$= \dots\dots\dots$$

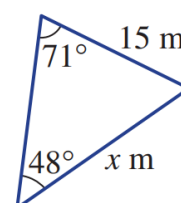
7.9

b



16.5

c

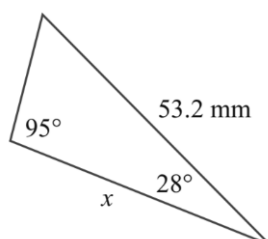


19.1

3 Calculate the side length x using sine rule to 1 d.p.

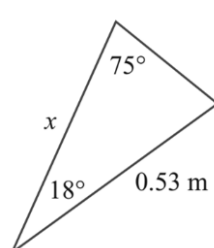
(Hint: Angle sum of a triangle is 180°)

a



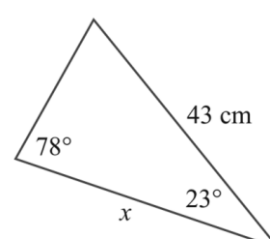
44.8 mm

b



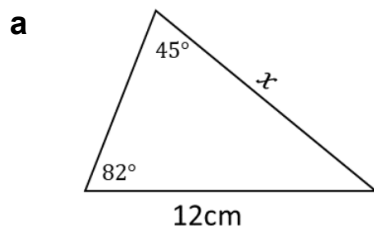
0.5 m

c

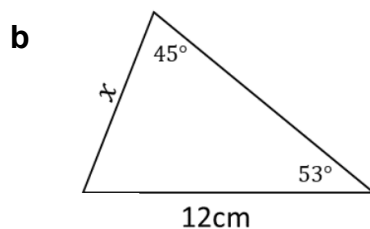


43.2 cm

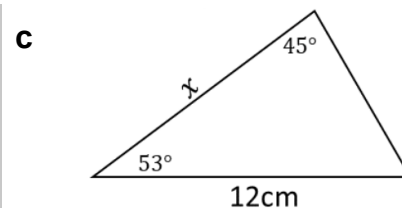
- 4 Calculate the side length x using sine rule. Round your answer to 1 d.p.



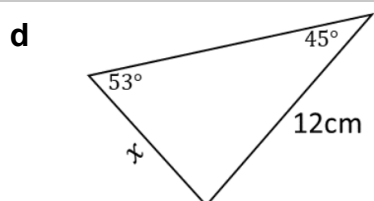
16.8 cm



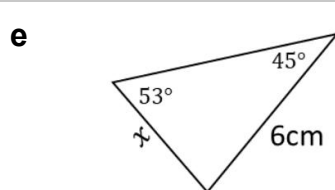
13.6 cm



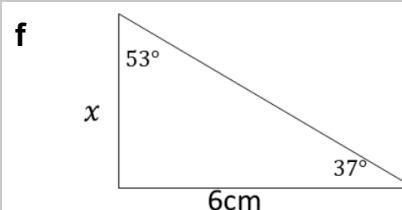
16.8 cm



10.6 cm

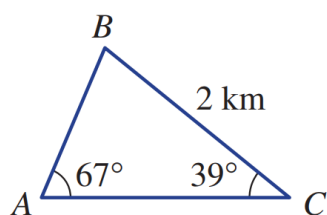


5.3 cm



4.5 cm

- 5 Three markers, A, B and C, map out the course for a cross-country race. Find the total distance of the race. Round your answer to 2 decimal places.



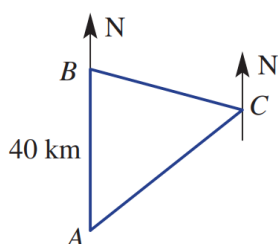
AB = 1.367 km, AC = 2.089 km, total length: 5.46 km

- 6 A ship heads due north from point A for 40 km to point B, and then heads on a bearing of 100° to point C. The bearing from C to A is 240° .

a Find $\angle ABC$.

b Hence find distance AC, to 1 d.p.

c Hence find distance BC, to 1 d.p.

 $\angle ABC = 80^\circ$

AC = 61.3 km

BC = 53.9 km

② Proof of the Sine Rule

Consider the non-right-angled triangle shown.

By looking at right-angled triangle $\triangle CPB$,

$$\sin B = \frac{h}{a}$$

$$\therefore h =$$

Similarly, from $\triangle CPA$,

$$\sin A = \frac{h}{b}$$

$$\therefore h =$$

By equating our two expressions for h :

$$a \sin B = b \sin A$$

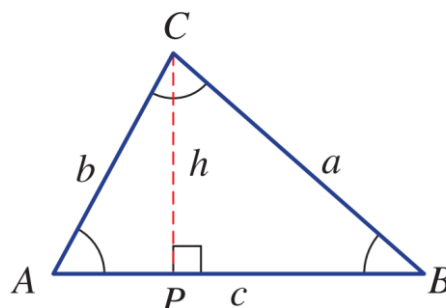
$$\therefore \frac{a}{\sin A} =$$

By constructing a perpendicular line from vertex B to AC and following the steps of the proof above, prove that:

$$\frac{a}{\sin A} = \frac{c}{\sin C}$$

Hence explain why:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



Let the perpendicular from B meet AC at point Q.

$$\text{In } \triangle CBQ, \sin C = \frac{BQ}{a}, \therefore BQ = a \sin C$$

$$\text{In } \triangle BAQ, \sin A = \frac{BQ}{c}, \therefore BQ = c \sin A$$

$$\therefore a \sin C = c \sin A$$

$$\frac{a}{\sin A} = \frac{c}{\sin C}$$

Since $\frac{a}{\sin A} = \frac{b}{\sin B}$ and $\frac{a}{\sin A} = \frac{c}{\sin C}$, all three are equal.

Sine Rule for Angles

Review

- ☐ Round to the nearest minute

a $34^{\circ} 40' 12''$

b $24^{\circ} 20' 34''$

c $37^{\circ} 49' 00''$

d $11^{\circ} 10' 49''$

- ☐ Solve trigonometric equations for an angle
Answer to the nearest minute

a $\sin \theta = 0.6248$

b $\sin \theta = \frac{5}{12}$

c $\sin \theta = \frac{4 \sin 60^{\circ}}{15}$

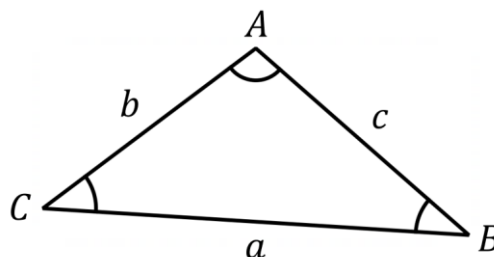
d $\frac{\sin \theta}{20} = \frac{\sin 25^{\circ}}{18}$

e $\frac{\sin 77^{\circ}}{12} = \frac{\sin \theta}{6}$

f $\frac{11}{\sin \theta} = \frac{17}{\sin 30^{\circ}}$

① Sine Rule (Finding an Angle)

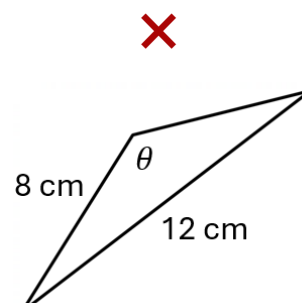
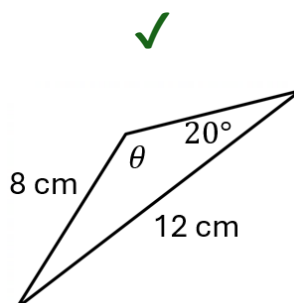
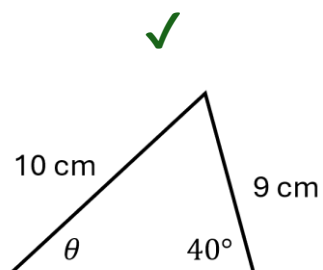
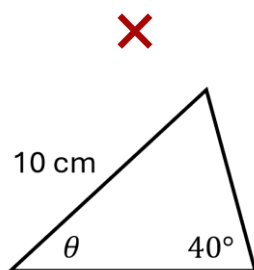
$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$



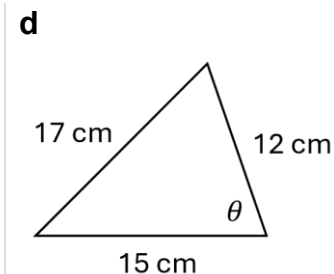
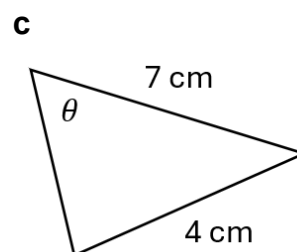
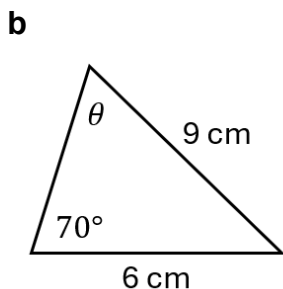
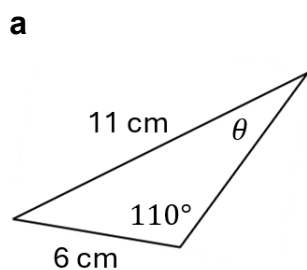
To use the sine rule to find an angle, you need:

- A complete 'opposite pair' – one side AND its opposite angle (both known).
- The opposite side to the angle you want to find.

② Identify whether you can use sine rule to find an angle



Identify whether you can use sine rule to find the missing angle.

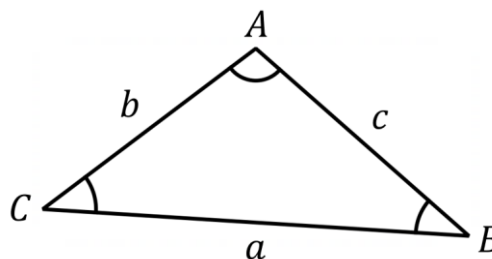


① Sine Rule (Finding an Angle)

When finding an angle, have sin in the numerator.

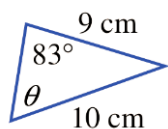
This makes it easier to solve for an angle.

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$



🔧 Example Calculate an angle using sine rule

Find the missing angle to the nearest minute.



$$\frac{\sin \theta}{9} = \frac{\sin 83^\circ}{10}$$

$$\times 9 \quad \times 9$$

$$\sin \theta = \frac{9 \sin 83^\circ}{10}$$

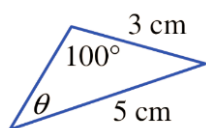
$$\theta = \sin^{-1} \left(\frac{9 \sin 83^\circ}{10} \right)$$

$$\approx 63^\circ 17'$$

✎ Guided Practice

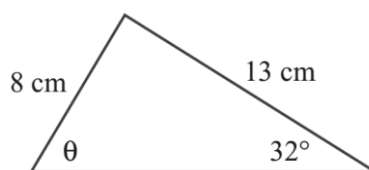
Find the missing angle to the nearest minute.

a



36° 13'

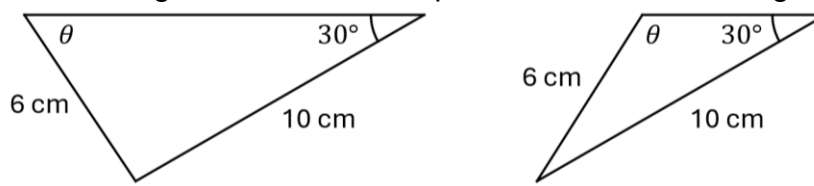
b



59° 27'

i Sine Rule (Finding an Obtuse Angle)

Given two sides and one angle, it is sometimes possible to make 2 triangles:



This is due to the identity: $\sin \theta = \sin(180^\circ - \theta)$

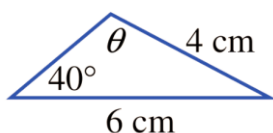
The calculator will always give the acute angle when using $\sin^{-1}(\quad)$

To get the obtuse angle, **subtract the result from 180**: $180^\circ - \theta$

You will learn the reason for this if you study Mathematics Advanced

Example Calculate an angle using sine rule

Find the missing angle to the nearest degree, given it is obtuse.



$$\frac{\sin \theta}{6} = \frac{\sin 40^\circ}{4}$$

$$\times 6 \quad \times 6$$

$$\sin \theta = \frac{6 \sin 40^\circ}{4}$$

$$\theta = \sin^{-1}\left(\frac{6 \sin 40^\circ}{4}\right)$$

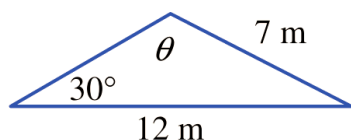
$$\approx 75^\circ$$

$$\therefore \theta = 180^\circ - 75^\circ = 105^\circ$$

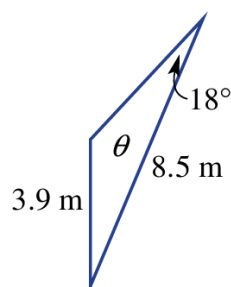
Guided Practice

Find the missing angle to the nearest degree, given it is obtuse.

a



b



1 Solve each equation for the unknown, correct to the nearest minute.

a

$$\frac{\sin 38^\circ}{4} = \frac{\sin \theta}{5}$$

50° 19'

b

$$\frac{\sin 51^\circ}{11} = \frac{\sin \theta}{9}$$

39° 29'

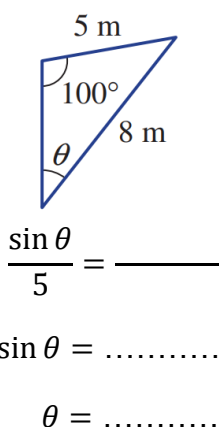
c

$$\frac{12}{\sin \theta} = \frac{18}{\sin 47^\circ}$$

29° 11'

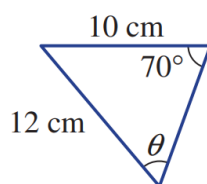
2 Find the value of the unknown in these triangles, correct to 1 decimal place.

a



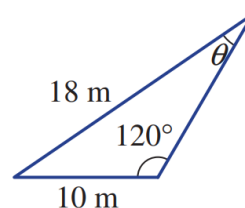
38.0°

b



51.5°

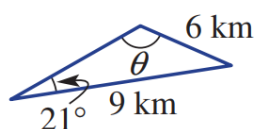
c



28.8°

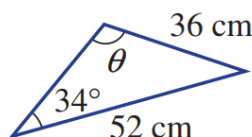
3 Calculate the unknown **obtuse** angle θ . Answer to 1 d.p.

a



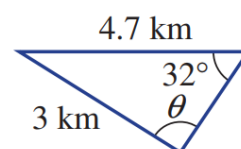
147.5°

b



126.1°

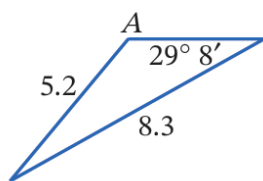
c



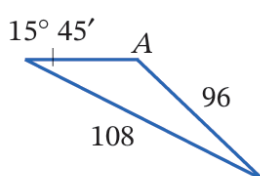
123.9°

- 4 Calculate the unknown **obtuse** angle $\angle A$. Answer to nearest minute.

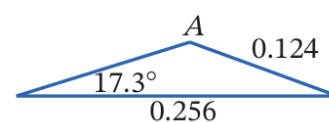
a

 $129^\circ 0'$

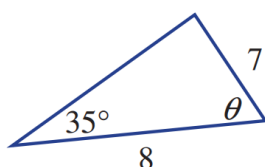
b

 $162^\circ 13'$

c

 $142^\circ 8'$

- 7 Ryan trying to solve this problem. He wrote $\frac{\sin 35^\circ}{7} = \frac{\sin \theta}{8}$.



- a Explain why he is incorrect.

 θ is not opposite the 8

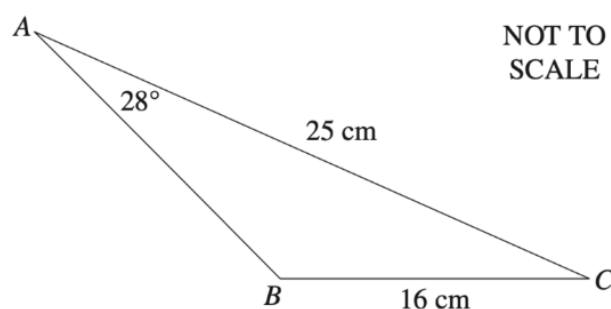
- b Solve for θ by finding the missing information first.

Find top angle using sine rule, $\phi = 40^\circ 58'$

$$\theta = 180 - 35 - 40^\circ 58' = 104^\circ 2'$$

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Find obtuse $\angle ABC$ to the nearest degree.



NOT TO SCALE

 133°

Cosine Rule for Sides

Review

- ☐ Solve quadratic equations
Answer to 1 d.p.

a

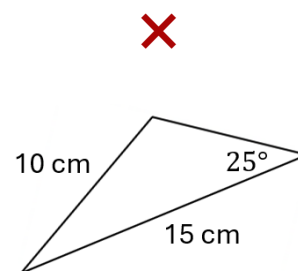
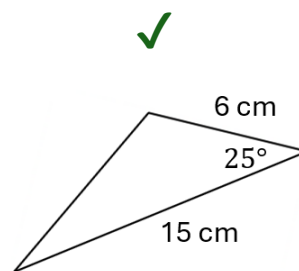
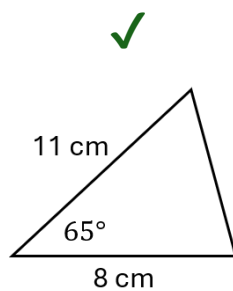
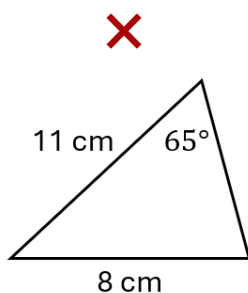
$$x^2 = 7^2 + 11^2 - 2(7)(11) \cos 60^\circ$$

b

$$x^2 = 8^2 + 15^2 - 2(8)(15) \cos 45^\circ$$

i Included Angle

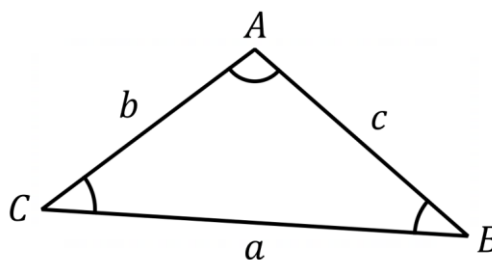
The **included angle** is the angle **between** two given sides.



① Cosine Rule (Finding a Side)

The cosine rule is Pythagoras' theorem, generalised for non-right-angled triangles.

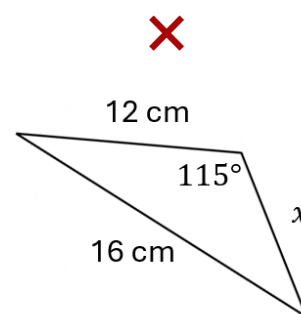
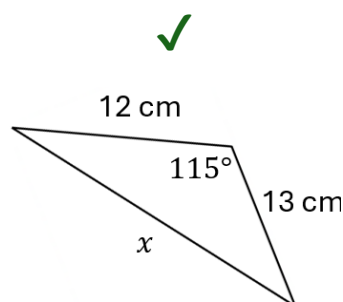
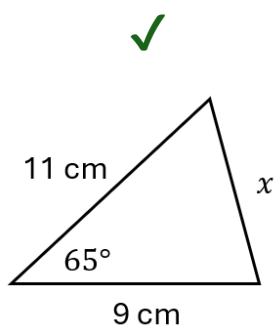
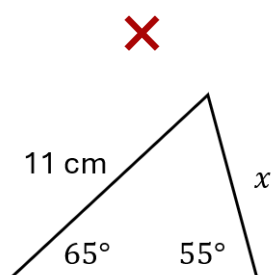
$$c^2 = a^2 + b^2 - 2ab \cos C$$



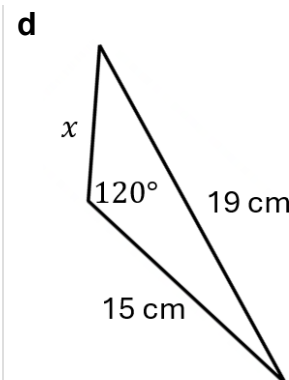
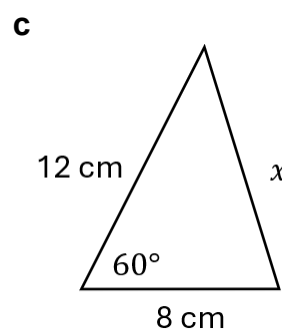
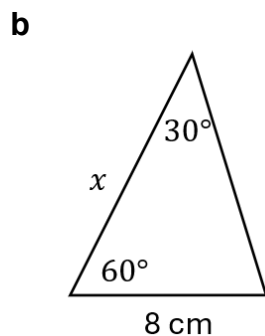
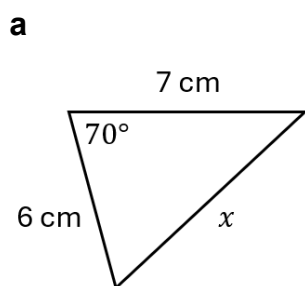
To use the cosine rule to find a side, you need:

- Two given sides.
- The included angle, C , which is opposite the side you want to find.

② Identify whether you can use cosine rule to find a side

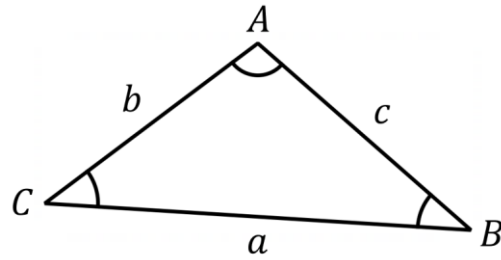


Identify whether you can use cosine rule to find the missing side.



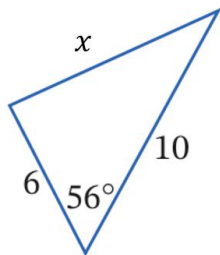
① Cosine Rule (Finding a Side)

$$c^2 = a^2 + b^2 - 2ab \cos C$$



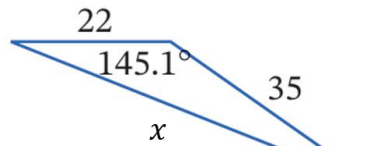
② Write Cosine rule equation from a diagram

a

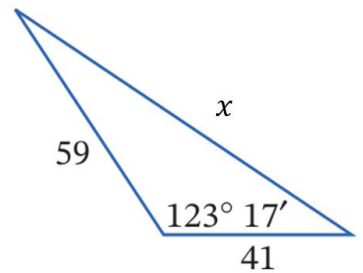


$$x^2 = 6^2 + 10^2 - 2(6)(10) \cos 56^\circ$$

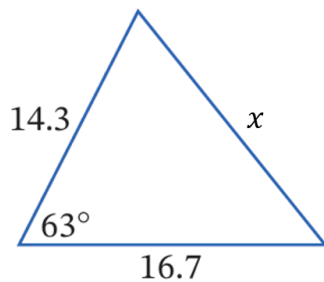
b



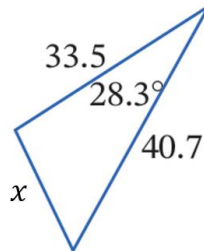
c



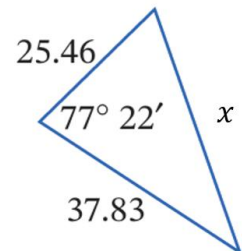
d



e

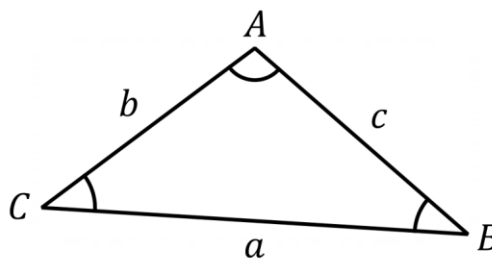


f



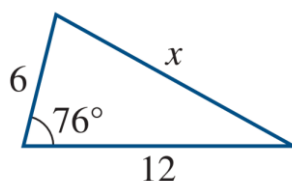
i Cosine Rule (Finding a Side)

$$c^2 = a^2 + b^2 - 2ab \cos C$$



Example Calculate a side using cosine rule

Calculate the length of x to 2 d.p.



$$c^2 = a^2 + b^2 - 2ab \cos C$$

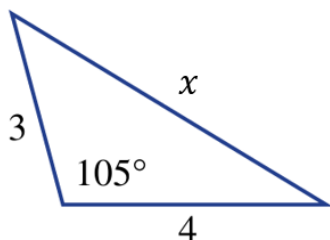
$$x^2 = 6^2 + 12^2 - 2(6)(12) \cos 76^\circ$$

$$x = \sqrt{6^2 + 12^2 - 2(6)(12) \cos 76^\circ}$$
$$\approx 12.05$$

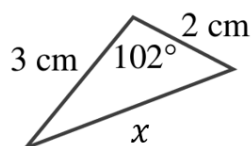
Guided Practice

Calculate the length of x to 2 d.p.

a



b



5.59

3.94

1 Solve for the unknown value in these equations. Answer to 2 d.p.

a $x^2 = 4^2 + 7^2 - 2(4)(7) \cos 120^\circ$

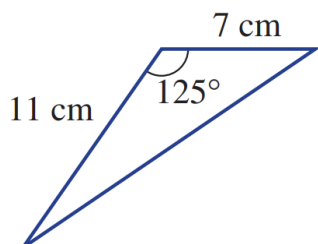
9.64

b $x^2 = 1.5^2 + 1.1^2 - 2(1.5)(1.1) \cos 70^\circ$

1.53

2 Find the length of the unknown side in these triangles, correct to 2 decimal places.

a



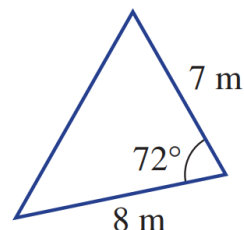
$$x^2 = (\quad)^2 + (\quad)^2 - 2(\quad)(\quad) \cos(\quad)$$

$$x = \sqrt{\quad}$$

$$x = \dots\dots\dots$$

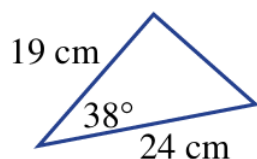
16.07 cm

b



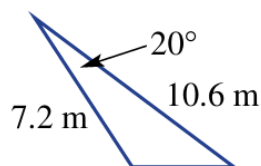
8.85 m

c



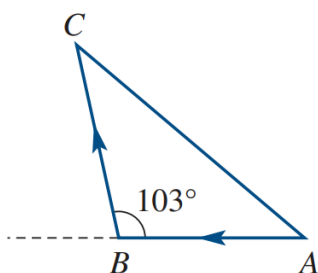
14.78 cm

d



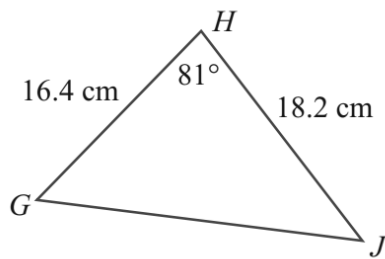
4.56 m

3 Ruby drives along a 14 km track from point A due west to a point B. She then turns and travels 19 km to point C. Calculate the distance from her starting point. Answer correct to one decimal place.



26.0 km

- 4 Calculate the perimeter of this triangle to 1 d.p.

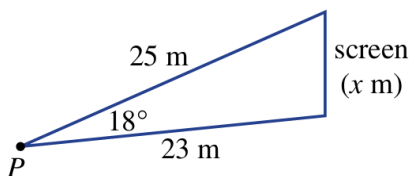


GJ = 22.5 cm, Perimeter = 57.1 cm

- 5 A triangular goat paddock has two sides of lengths 320 m and 170 m, and a 71° angle between them. Find the length of the third side, correct to the nearest metre.

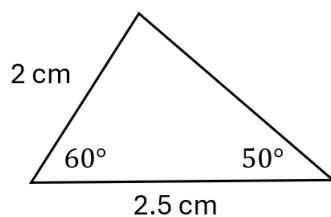
310 m

- 6 The viewing angle to a vertical screen is 18° and the distances between the viewing point, P, and the top and bottom of the screen are 25 m and 23 m, respectively. Find the height of the screen (x m), correct to the nearest centimetre.



7.76 m (or 776 cm)

- 7 Find the length of the missing side to 1 d.p. by using cosine rule, then check your answer using sine rule.



Cosine rule:

$$c^2 = 2^2 + 2.5^2 - 2(2)(2.5) \cos 60^\circ$$

$$c = 2.3 \text{ cm}$$

Sine rule:

$$\frac{c}{\sin 60^\circ} = \frac{2}{\sin 50^\circ}$$

$$c = \frac{2 \sin 60^\circ}{\sin 50^\circ}$$

$$c = 2.3 \text{ cm}$$

② Proof of the Cosine Rule

Consider the non-right-angled triangle shown.

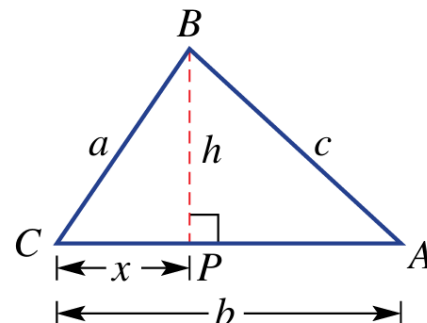
Write an expression for length AP

Use Pythagoras' Theorem in $\triangle CBP$ to write an equation relating c , x , and h

Use Pythagoras' Theorem in $\triangle ABP$ to write an equation relating b , a , x , and h

$$b - x$$

$$x^2 + h^2 = a^2$$



$$(b - x)^2 + h^2 = c^2$$

By making h^2 the subject of your two equations above, eliminate h and simplify your result.

$$h^2 = a^2 - x^2 \text{ and } h^2 = c^2 - (b - x)^2$$

$$\begin{aligned} a^2 - x^2 &= c^2 - (b - x)^2 \\ a^2 - x^2 &= c^2 - (b^2 - 2bx + x^2) \\ a^2 - x^2 &= c^2 - b^2 + 2bx - x^2 \\ a^2 &= c^2 - b^2 + 2bx \end{aligned}$$

Use $\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$ to write an expression for $\cos C$.

$$\cos C = \frac{x}{a}$$

Combine your equations from the two parts above to prove cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\begin{aligned} x &= a \cos C \\ a^2 &= c^2 - b^2 + 2ab \cos C \\ \therefore c^2 &= a^2 + b^2 - 2ab \cos C \end{aligned}$$

Cosine Rule for Angles

Review

- ☐ Solve trigonometric equations for an angle
Answer to the nearest minute

a

$$\cos \theta = \frac{12^2 + 5^2 - 8^2}{2(12)(5)}$$

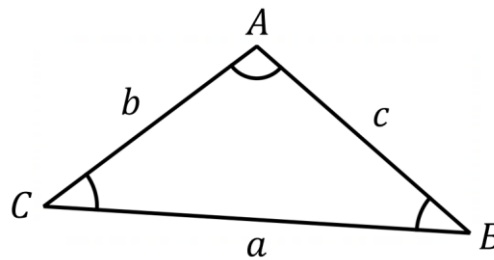
b

$$\cos \theta = \frac{11^2 + 5^2 - 7^2}{2(11)(5)}$$

i Cosine Rule

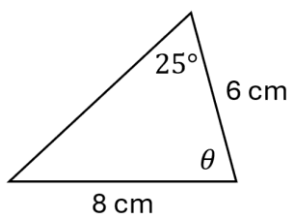
We can rearrange the cosine rule to find an angle.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

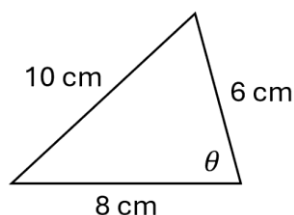


To use the cosine rule to find an angle, you need the lengths of all three sides.

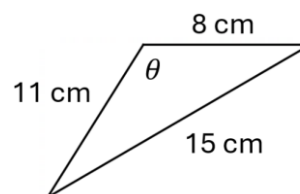
✗



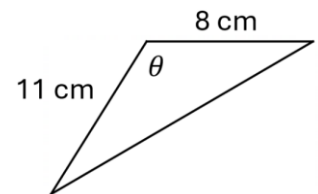
✓



✓



✗

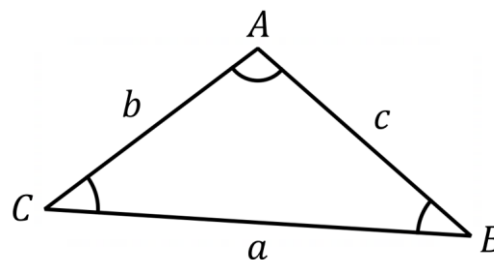


① Cosine Rule (Finding an Angle)

c is always the side OPPOSITE the angle you want to find.

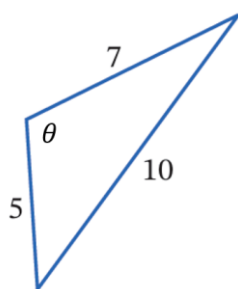
The other two sides are a and b (order doesn't matter).

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



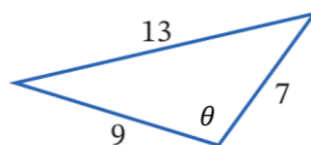
② Write Cosine rule equation from a diagram

a

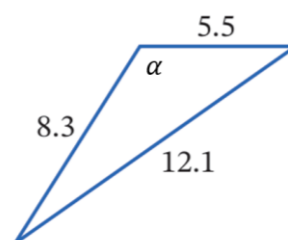


$$\cos \theta = \frac{5^2 + 7^2 - 10^2}{2(5)(7)}$$

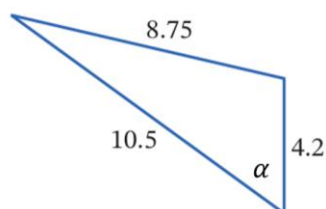
b



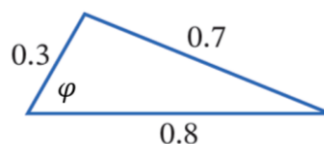
c



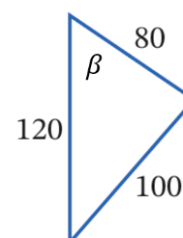
d



e

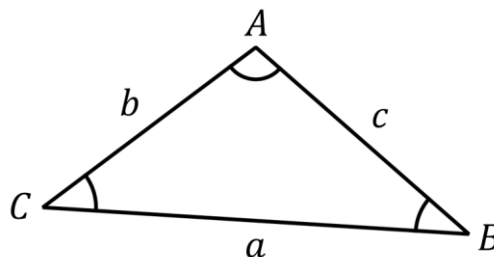


f



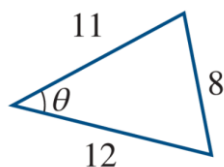
① Cosine Rule (Finding an Angle)

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



🔧 Example Calculate an angle using cosine rule

Calculate θ to the nearest minute.



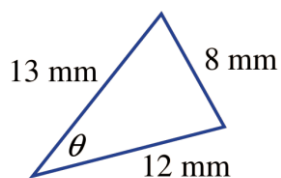
$$\cos \theta = \frac{11^2 + 12^2 - 8^2}{2(11)(12)}$$

$$\theta = \cos^{-1} \left(\frac{11^2 + 12^2 - 8^2}{2(11)(12)} \right)$$

$$\theta = 40^\circ 25'$$

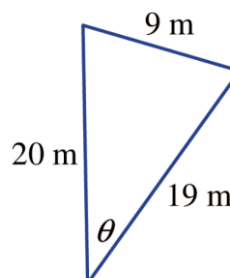
📝 Guided Practice

a



37° 3'

b



26° 32'

1 Solve for the unknown value in these equations. Answer to the nearest minute.

a

$$\cos \theta = \frac{7^2 + 6^2 - 9^2}{2(7)(6)}$$

87°16'

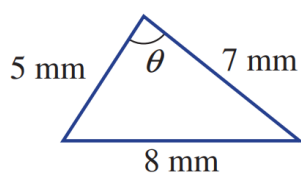
b

$$\cos \theta = \frac{21^2 + 30^2 - 18^2}{2(21)(30)}$$

36°11'

2 Find the value of the unknown in these triangles, correct to 2 decimal places.

a



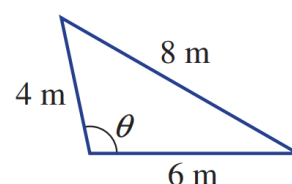
$$\cos \theta = \frac{(\quad)^2 + (\quad)^2 - (\quad)^2}{2(\quad)(\quad)}$$

$$\theta = \cos^{-1} \left(\frac{(\quad)^2 + (\quad)^2 - (\quad)^2}{2(\quad)(\quad)} \right)$$

$\theta = \dots\dots\dots$

81.79°

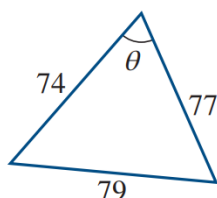
b



104.48°

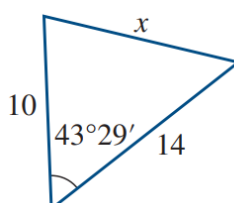
3 Calculate the unknowns. Answer to 2 d.p. for lengths and the nearest degree for angles.

a



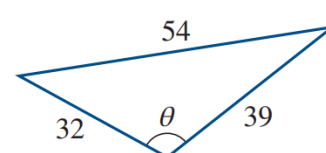
63°

b



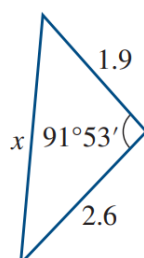
9.64

c



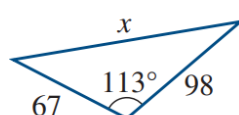
99°

d



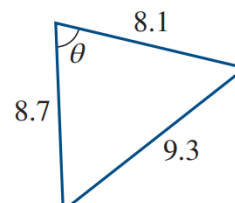
3.27

e



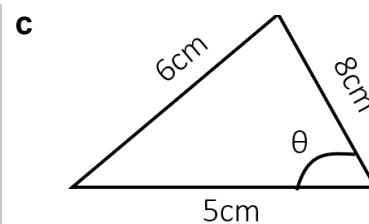
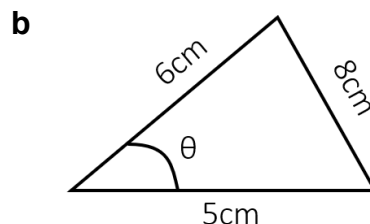
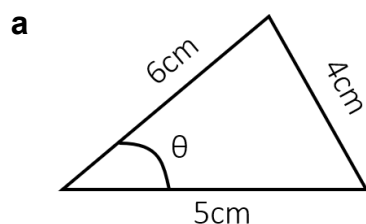
138.65

f



67°

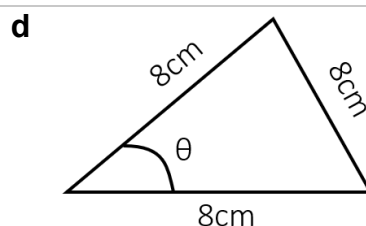
- 4 Calculate the angle θ using cosine rule. Answer to 2 d.p.



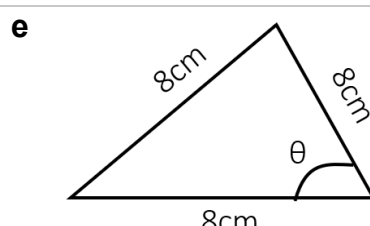
41.40°

92.87°

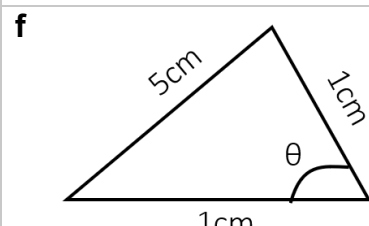
48.51°



60°



60°



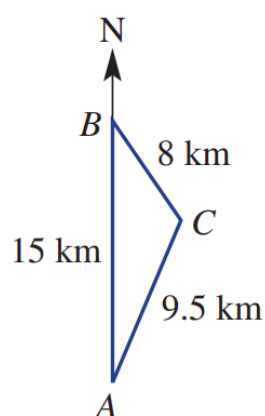
impossible triangle

- 5 A triangle has side lengths 10 m, 7 m, and 13 m. Find the largest angle, to one decimal place.

Largest angle always opposite longest side

98.2°

- 6 Three camp sites, A, B and C, are shown.
If campsite B is due north of camp site A, find the following, correct to 1 decimal place.



- a** The bearing of campsite C from campsite B

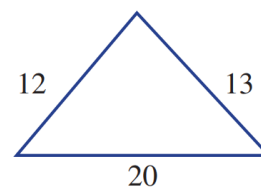
145.9°

- b** The bearing of campsite A from campsite C

208.2°

7 Consider the triangle shown. Answer to the nearest degree.

a Use cosine rule to find the size of the largest angle.

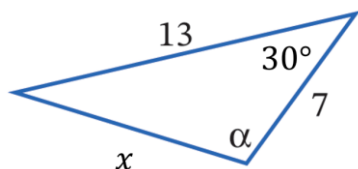


Largest angle opposite longest side: 106°

b Use sine rule to find the size of the smallest angle.

Smallest angle opposite shortest side: 35°

8 Consider the triangle below.



a Find x using cosine rule to 4 d.p.

7.7707

b Hence find α to 2 d.p using cosine rule.

$$\alpha = \cos^{-1} \left(\frac{7.7707^2 + 7^2 - 13^2}{2(7.7707)(7^2)} \right) = 123^\circ 14'$$

c Use sine rule to find α to check your answer.

$$\alpha = \sin^{-1} \left(\frac{13 \sin 30}{7.77} \right) = 56^\circ 46'$$

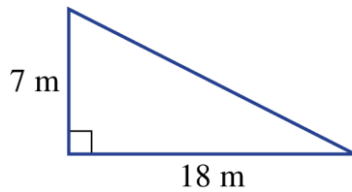
Obtuse angle, so $180 - 56^\circ 46' = 123^\circ 14'$

Area of a Triangle using Trigonometry

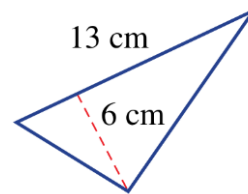
Review

- ☐ Calculate area of a triangle, given perpendicular dimensions

a

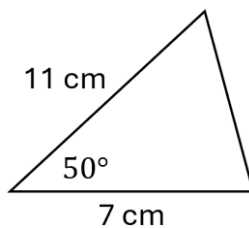


b

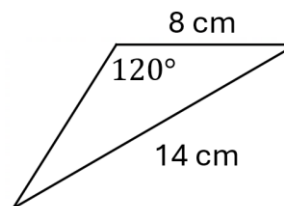


- ☐ Identify included angle.

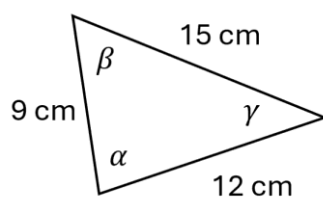
- a** Is 50° the included angle between 11 and 7?



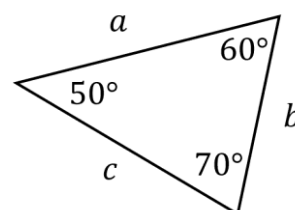
- b** Is 120° the included angle between 8 and 14 cm?



- c** What is the angle included between 12 cm and 15 cm?



- d** What two sides make the included angle of 50° ?

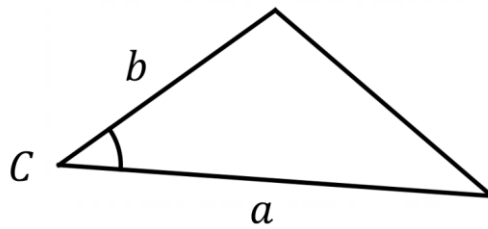


i SAS Area Rule

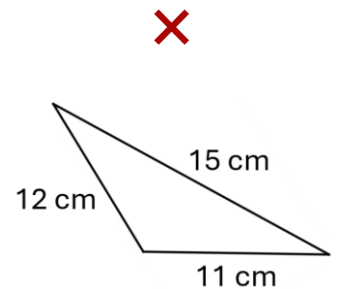
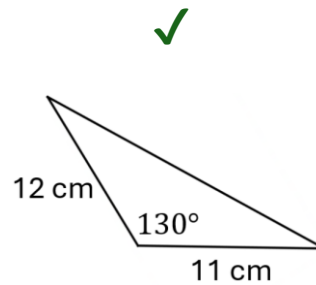
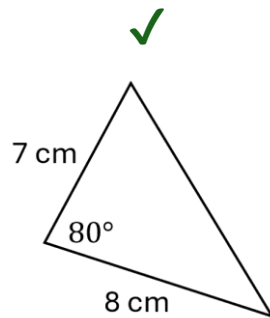
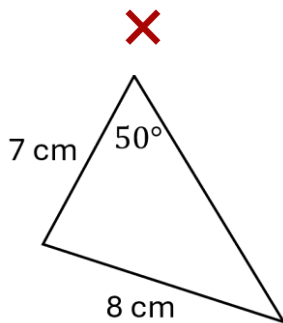
$$A = \frac{1}{2}ab \sin C$$

Where:

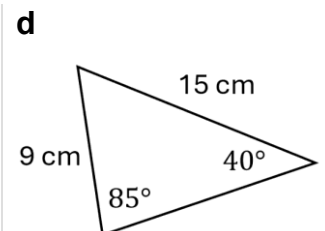
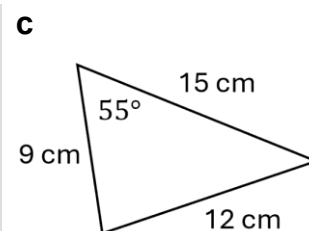
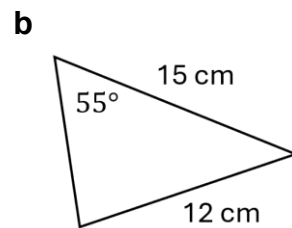
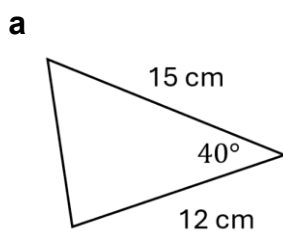
- a and b are two sides.
- C is the included angle.



? Identify whether you can use SAS area rule to find area

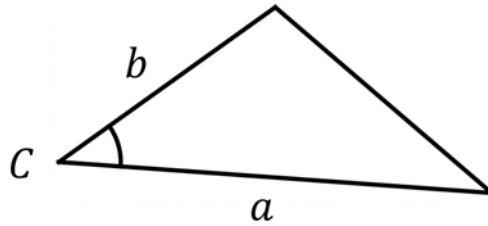


Identify whether you can use the SAS area rule to find the missing side.



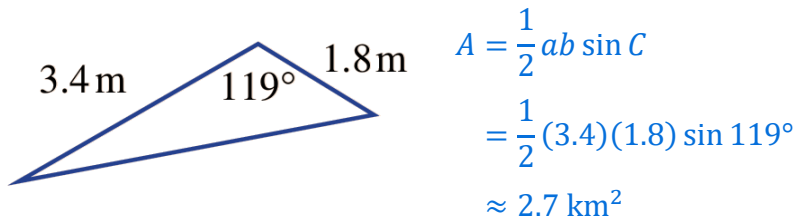
SAS Area Rule

$$A = \frac{1}{2}ab \sin C$$



Example Calculate area of a triangle using SAS area rule

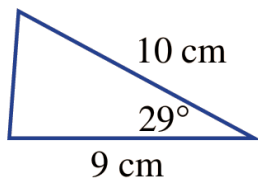
Find the area of this triangle, to 1 d.p.



Guided Practice

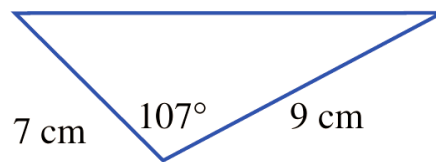
Find the area of these triangles, to 1 d.p.

a



21.8 cm²

b

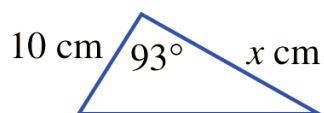


30.1 cm²

Find the side length, to 2 d.p., given the area.

a

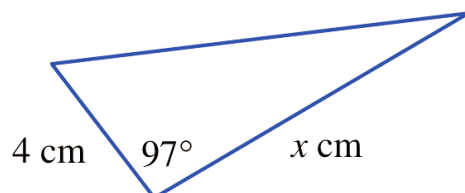
$$A = 70 \text{ cm}^2$$



14.02 cm

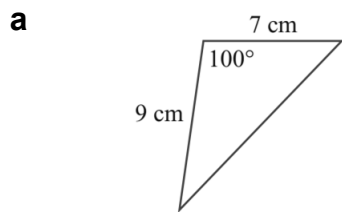
b

$$A = 37 \text{ cm}^2$$



18.64 cm

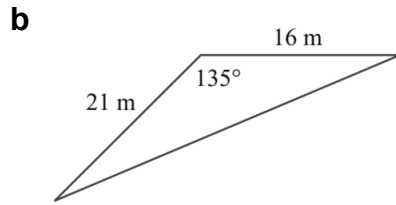
1 Calculate the area of each triangle, correct to one decimal place.



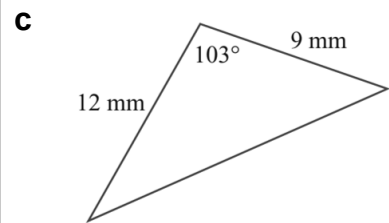
$$A = \frac{1}{2} (\quad) (\quad) \sin(\quad)$$

$$= \dots\dots\dots$$

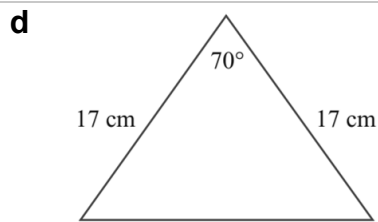
31.0 cm²



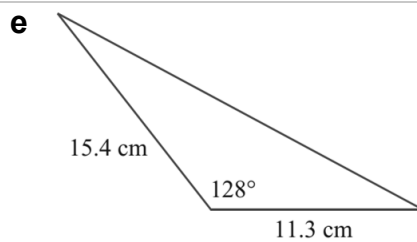
118.8 m²



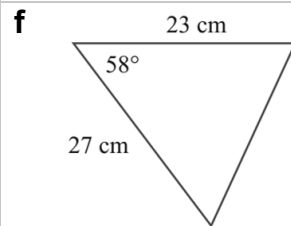
52.6 mm²



135.8 cm²

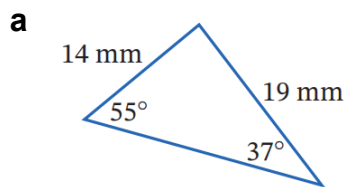


68.6 cm²

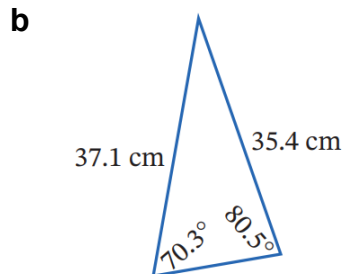


263.3 cm²

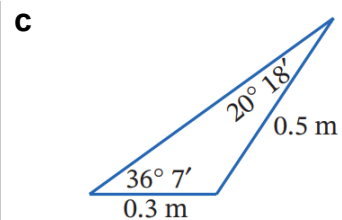
2 Find the area of each triangle. Answer to 1 d.p.



132.9 mm²

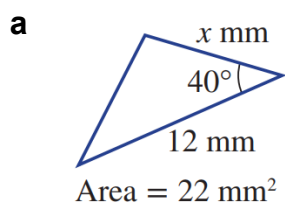


320.4 cm²

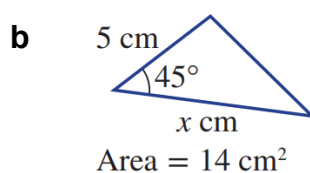


0.1 m²

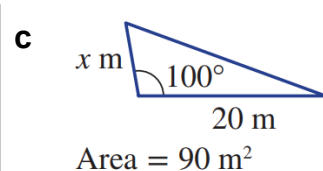
3 Calculate the value of x correct to one decimal place.



$x = 5.7$



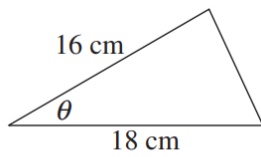
$x = 7.9$



$x = 9.1$

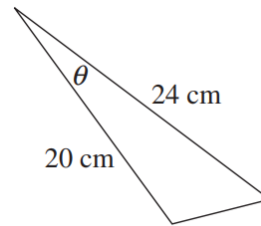
4 Calculate the value of θ given each triangle has area 72 cm^2 . Answer to the nearest minute.

a



30°

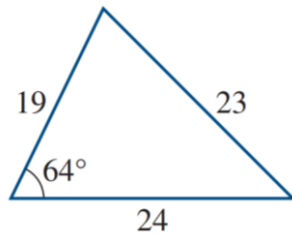
b



$17^\circ 27'$

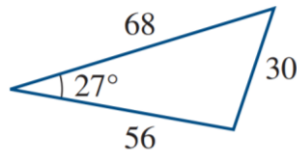
5 Find the area of these triangles.

a



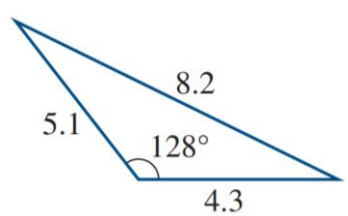
204.9 sq units

b



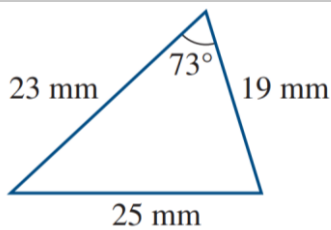
864.4 sq units

c



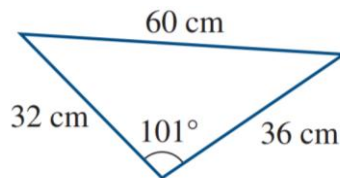
8.6 sq units

d



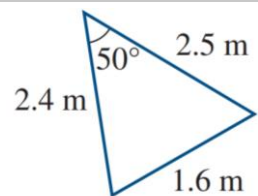
209.0 mm^2

e



565.4 cm^2

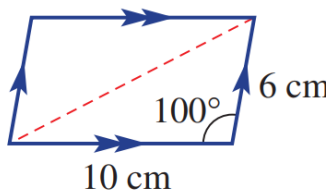
f



2.3 m^2

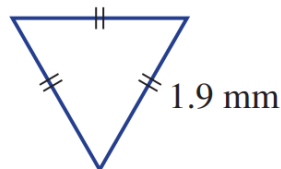
6 Find the area of these shapes.

a



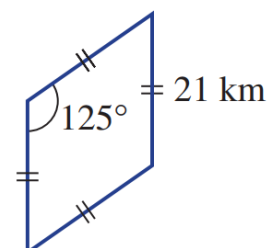
59.09 cm^2

b



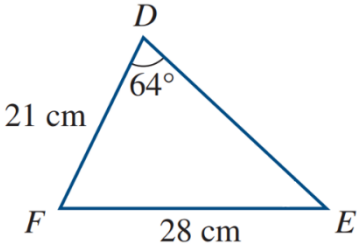
1.56 mm^2

c



361.25 km^2

7 Consider the triangle.



a Explain why we cannot find the area of the triangle.

The angle is not in between two sides.

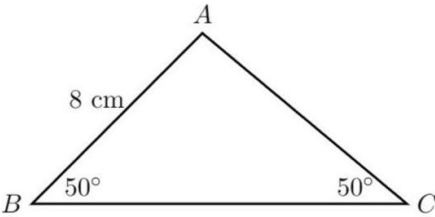
b Using sine rule, find $\angle DEF$ to the nearest degree.

42°

c Hence calculate the area to the nearest square cm.

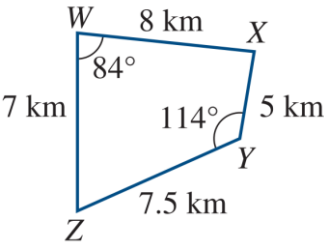
282 cm^2

8 Calculate the area of this triangle to 1 d.p.



31.5 cm^2

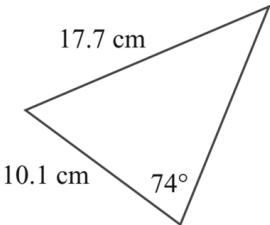
9 A drawing of a farmer's quadrilateral property is shown. By considering $\triangle WXZ$ and $\triangle XYZ$, calculate the area of the property in square km correct to one decimal place.



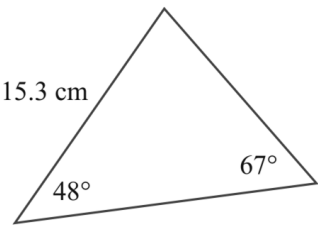
45.0 km^2

10 Calculate the area of these triangles. Answer in square cm to 1 d.p.
(Hint: Use sine rule and angle sum of a triangle)

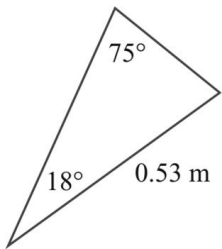
a



b



c

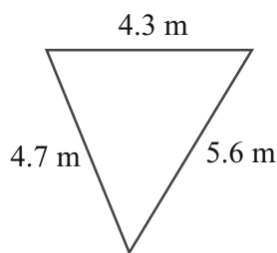


85.4 cm^2

85.6 cm^2

448.7 cm^2

- 11 Use cosine rule to find an angle, and hence calculate the area of this triangle to 1 d.p.



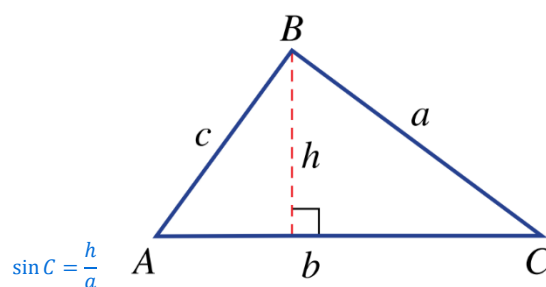
9.8 m²

② **Proof** of the SAS area formula

Consider the non-right-angled triangle shown.

The area of a triangle (from Year 8) is $A = \frac{1}{2}bh$

Write the equation for $\sin C$



$$\sin C = \frac{h}{a}$$

Make h the subject of the equation you wrote above.

$$h = a \sin C$$

By combining $A = \frac{1}{2}bh$ and $h = a \sin C$, derive the side-angle-side formula for triangle area.

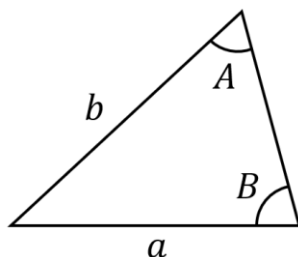
$$\begin{aligned} A &= \frac{1}{2}b(a \sin C) \\ &= \frac{1}{2}ab \sin C \end{aligned}$$

Selecting an Appropriate Method

① Methods for Non-Right-Angled Triangles

For non-right-angled triangles, determine what is involved in the question:

2 sides 2 angles

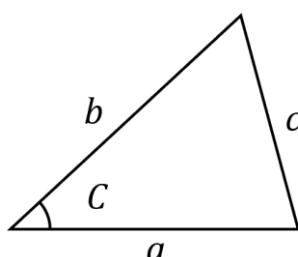


Sine Rule

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

3 sides 1 angle

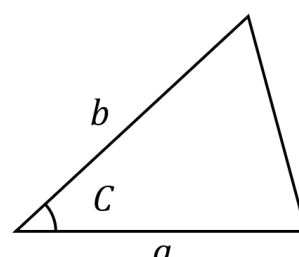


Cosine Rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Area

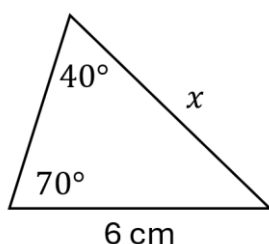


SAS Area Rule

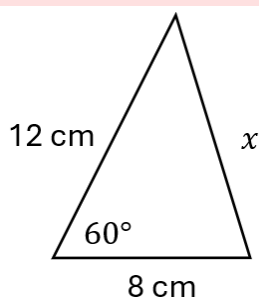
$$A = \frac{1}{2} ab \sin C$$

② Identify which rule you would use to find the unknown

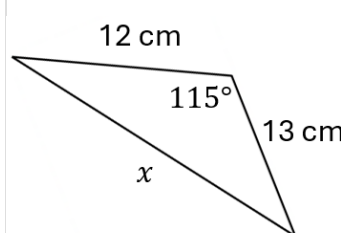
a



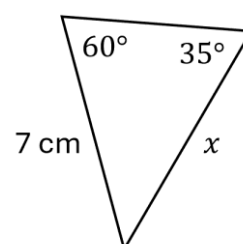
b



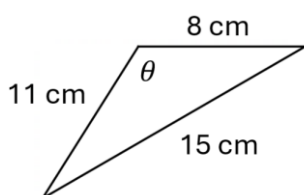
c



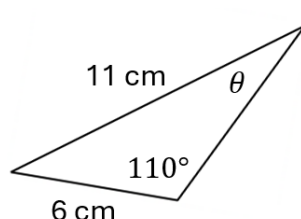
d



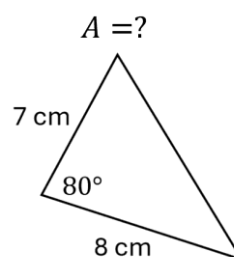
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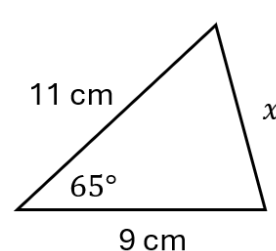
f



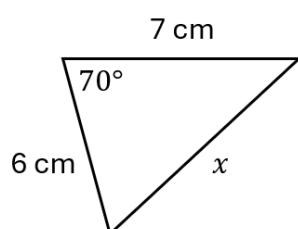
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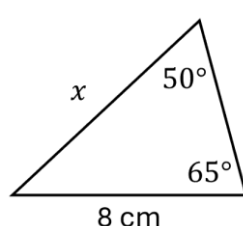
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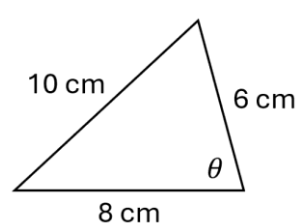
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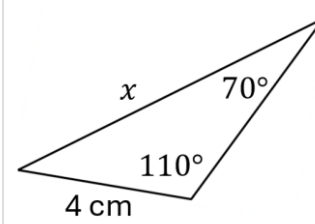
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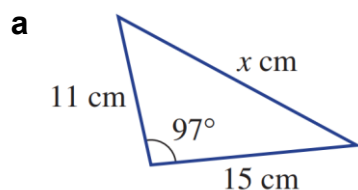
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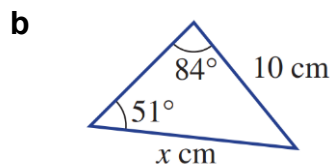
l



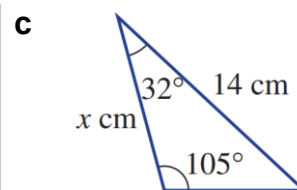
1 Decide whether cosine rule or sine rule should be used to find x . Do not solve for x .



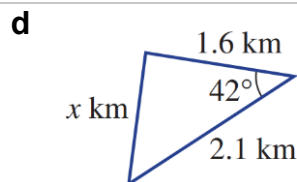
cosine



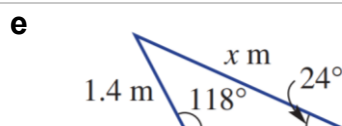
sine



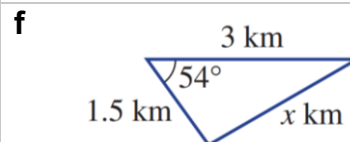
sine



cosine

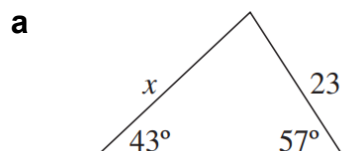


sine

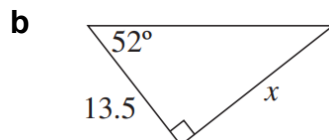


cosine

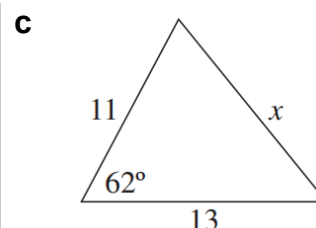
2 Use an appropriate method to find the unknown. Give lengths to 1 d.p. and angles to nearest degree.



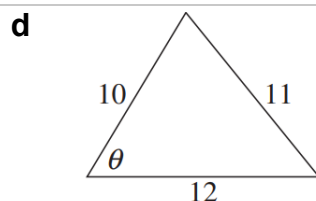
28.3



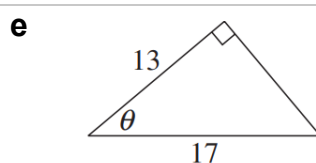
17.3



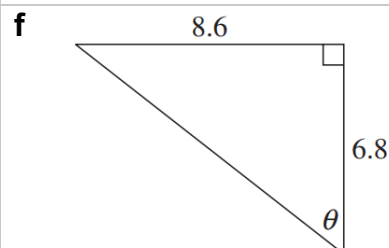
12.5



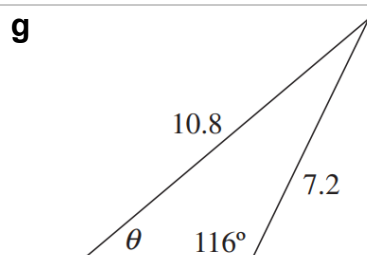
59°



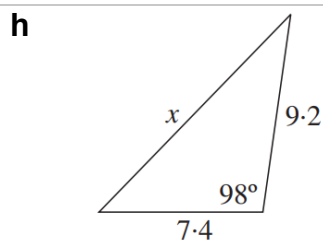
40°



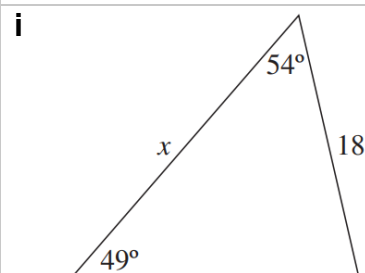
52°



37°



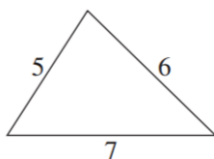
12.6



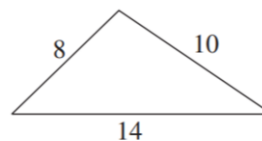
23.2

3

- a** Find the smallest angle, correct to the nearest minute.

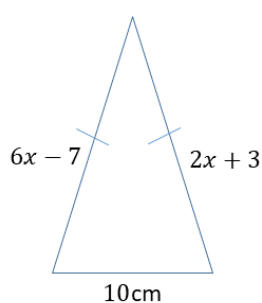
 $44^{\circ}25'$

- b** Find the largest angle, correct to the nearest minute.

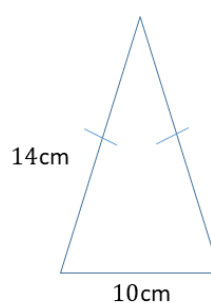
 $101^{\circ}32'$

4

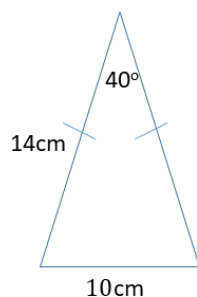
- a** Calculate the perimeter.

 26 cm

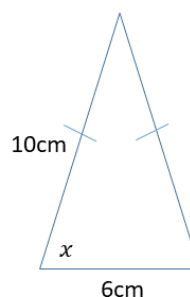
- b** Calculate the height in exact form.

 $\sqrt{171} \text{ cm}$

- c** Calculate the area to the nearest whole.

 63 cm

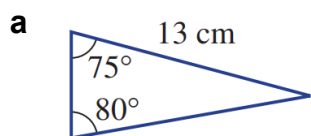
- d** Calculate the angle to 1 d.p.

 72.5°

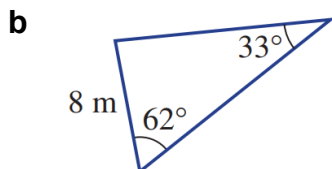
- 5** A triangle has sides of 21 m, 17 m, and 10 m.
Find the size of the largest angle, correct to the nearest degree.

 99°

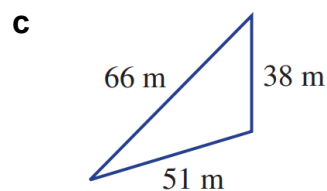
- 6 Calculate the area of these triangles by using either sine rule or cosine rule to find the required information. Give your answer to 2 d.p.



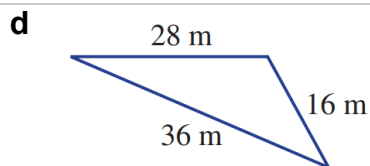
35.03 cm²



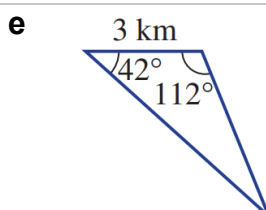
51.68 m²



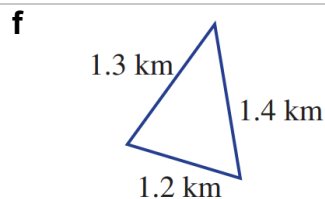
965.88 m²



214.66 m²

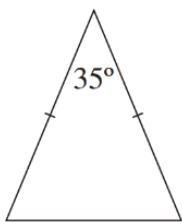


6.37 km²



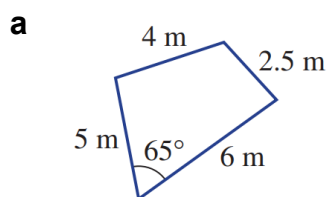
0.72 km²

- 7 The area of this isosceles triangle is 35 cm². Find the length of the equal sides to 2 d.p.

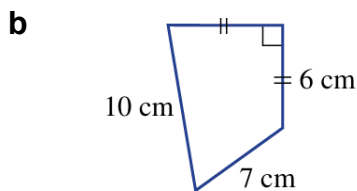


11.05 cm

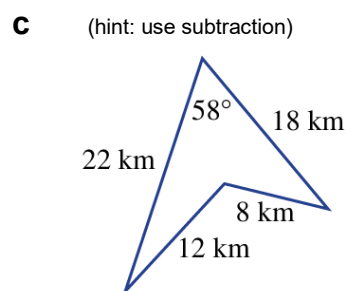
- 8 Calculate the area of these shapes. Round to 1 d.p.



17.3 m²



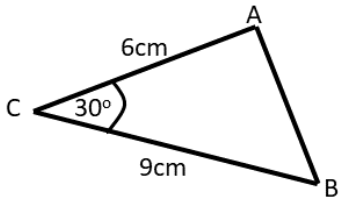
48 cm²



124.8 km²

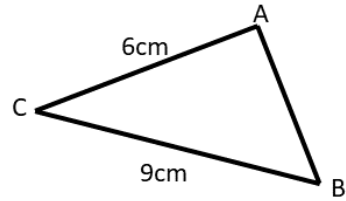
9

c Find the length of AB to 2 d.p.



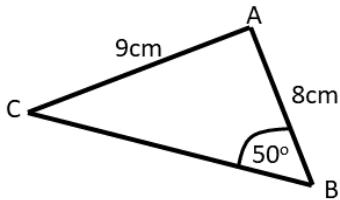
4.84 cm

d The area of the triangle is 9.23 cm^2 . Find the size of $\angle ACB$ to the nearest degree.



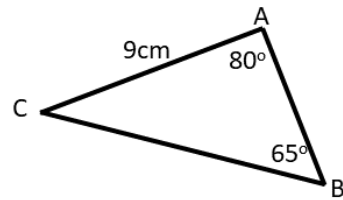
20°

e Find the size of angle $\angle BAC$ to 1 d.p.



87.1°

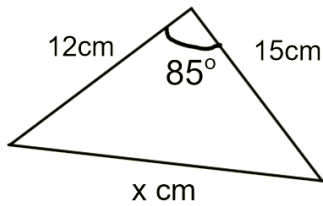
f Find the perimeter of the triangle to 2 d.p.



24.48 cm

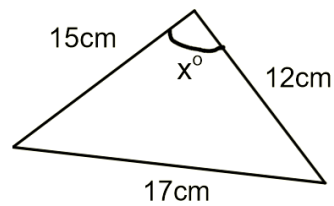
10 Find the unknown value, round lengths to 2 d.p. and angles to the nearest degree.

a



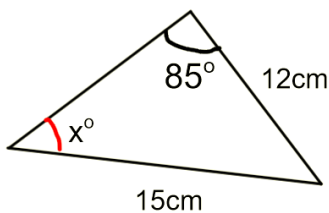
18.37 cm

b



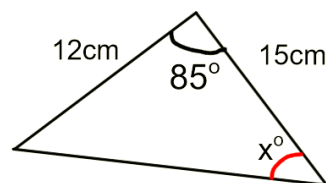
77°

c



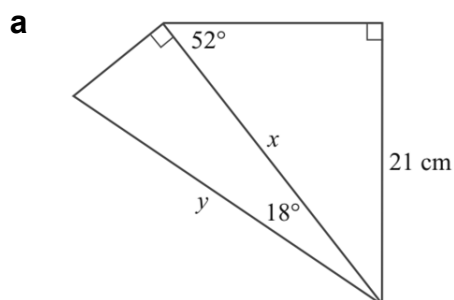
53°

d

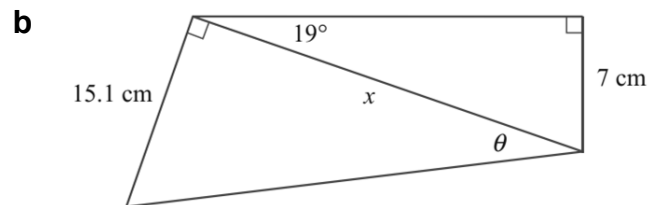


41°

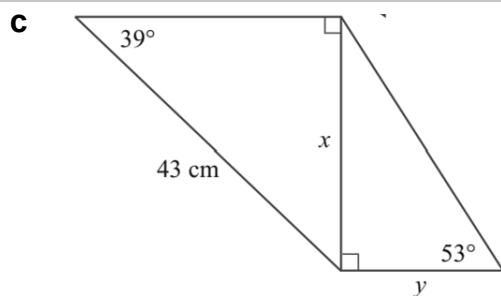
11 Determine the value of the variables to one d.p. for lengths and the nearest minute for angles.



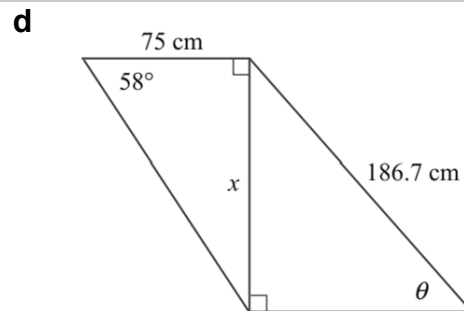
$$x = 26.6 \text{ cm}, y = 28.0 \text{ cm}$$



$$x = 21.5 \text{ cm}, \theta = 35^\circ 5'$$



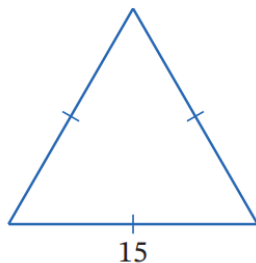
$$x = 27.1 \text{ cm}, y = 20.4 \text{ cm}$$



$$x = 120.0 \text{ cm}, \theta = 40^\circ 0'$$

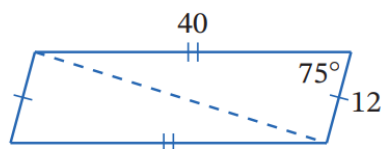
12 Calculate the area of each shape correct to one decimal place. All measurements are in metres.

a Equilateral triangle



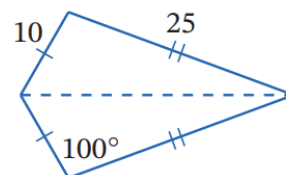
$$97.4 \text{ m}^2$$

b Parallelogram



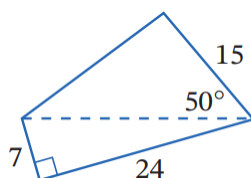
$$463.6 \text{ m}^2$$

c Kite



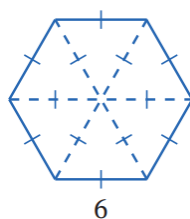
$$246.2 \text{ m}^2$$

d Quadrilateral



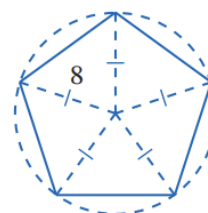
$$227.6 \text{ m}^2$$

e Regular hexagon



$$93.5 \text{ m}^2$$

f Regular pentagon inscribed in a circle

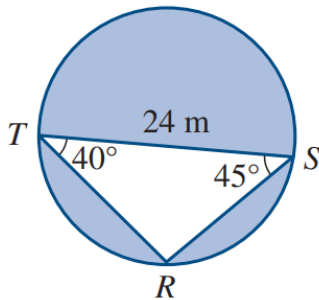


$$152.2 \text{ m}^2$$

Mixed Multistep Trigonometry Problems

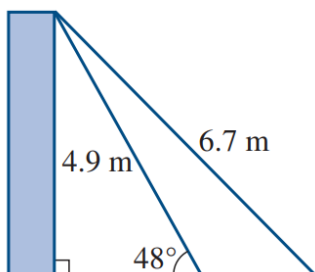
FOUNDATION

- 1 Andrew plans to build a triangular flower bed in the lower part of a circular garden. The length of TS is 24m, $\angle TSR$ is 45° and $\angle RTS$ is 40° .
- What is the size of $\angle SRT$?
 - Find the length of TR . Answer correct to the nearest metre.



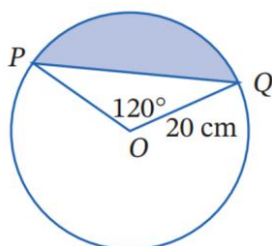
a) 95° b) 17 m

- 2 Two ladders are the same distance up the wall. The shorter ladder is 4.9m long and makes an angle of 48° with the ground. The longer ladder is 6.7m long.
- How far up the wall do the ladders reach?
Answer correct to one decimal place.
 - What is the angle, to the nearest degree, that the longer ladder makes with the ground?
 - Find the distance between the ladders on the ground. Answer to one decimal place.



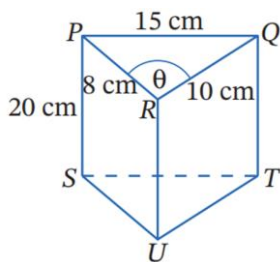
a) 3.6 m, b) 33° c) 2.3 m

- 3 O is the centre of a circle of radius 20 cm. Calculate, correct to one decimal place, the area of:
- Sector OPQ .
 - Triangle OPQ .
 - The shaded segment.



a) 418.9 cm^2 b) 173.2 cm^2 c) 245.7 cm^2

- 4 A triangular prism has base edge of 8 cm, 10 cm, and 15 cm, and a height of 20 cm.
- Calculate the size of $\angle PRQ$, correct to the nearest degree.
 - Find the area of $\triangle PQR$, correct to the nearest cm^2 .
 - Find the volume of the prism.

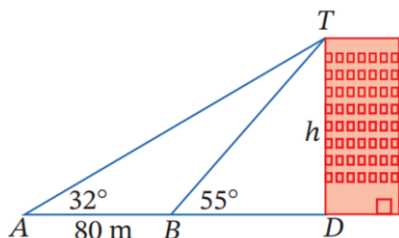


a) 112° b) 37 cm^2 c) 740 cm^3

- 5 The area of an equilateral triangle is 600 cm^2 .
Calculate the perimeter of the triangle, correct to 1 d.p.

111.7 cm

- 6 The angles of elevation of a building measured from 2 positions 80 m apart of 32° and 55° .
- Explain why $\angle ATB = 23^\circ$
 - Find, correct to 2 decimal places, the length of BT .
 - Hence find the height h of the building, correct to the nearest metre.

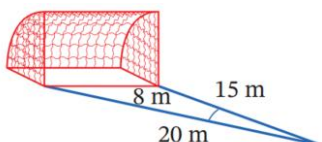


a) $\angle ATB = 55^\circ - 32^\circ$ (exterior angle theorem), other answers possible

b) 108.50 m

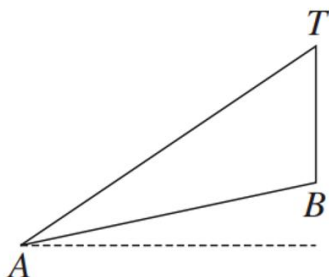
c) 89 m

- 7 A soccer goal is 8 m wide. A player shoots for goal (along the ground) when 20 m from one post and 15 m from the other post. Within what angle (correct to the nearest 0.1 degree) must the shot be made for the player to have a chance of scoring a goal?



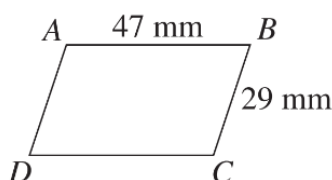
20.8°

- 8 Melissa is standing at A on a path that leads to the base B of a vertical flagpole. The path is inclined at 12° to the horizontal, and the angle of elevation of the top T of the flagpole from A is 34° .
- Explain why $\angle TAB = 22^\circ$ and $\angle ABT = 102^\circ$
 - Given that $AB = 20$ m, find the height of the flagpole, correct to the nearest metre.

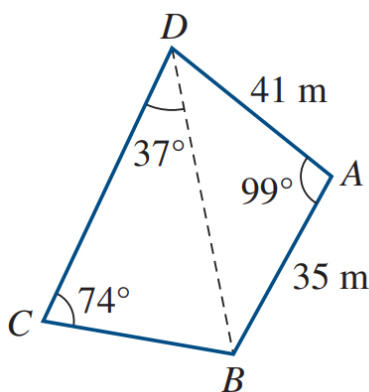


9 metres

- 9 A parallelogram $ABCD$ has sides $AB = DC = 47$ mm and $AD = BC = 29$ mm. The longer diagonal BD is 60 mm. Find the following, round your answer to the nearest minute.
- Use the cosine rule to find the size of $\angle BCD$.
 - Use co-interior angles to find the size of $\angle ABC$.

a) $101^\circ 38'$ b) $78^\circ 22'$

- 10 A survey of a park is shown in the diagram. A path is proposed from B to D and a fence is required from B to C to D .
- What is the length of the path? Answer correct to the nearest metre.
 - Calculate the required length of fencing. Answer correct to the nearest metre.
 - What is the area of the park? Answer correct to the nearest square metre.



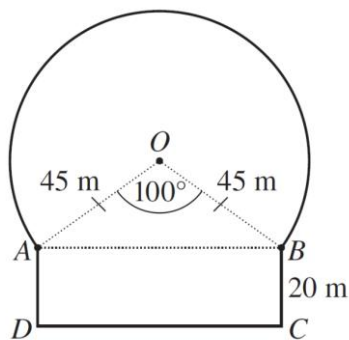
a) 58 m

b) 93 m

c) 1689 m^2

11 2016 HSC Standard 2 Band 4

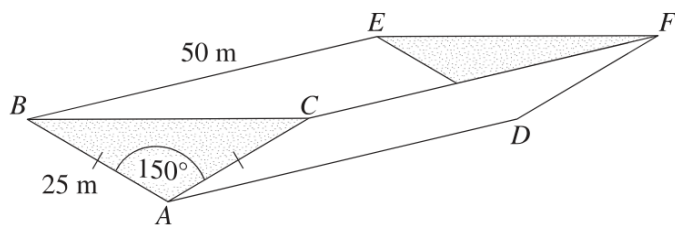
A school playground consists of part of a circle and a rectangle as shown in the diagram.
What is the area of the whole playground, correct to the nearest square metre?



6971 m²

12 2022 HSC Standard 2 Band 5

A dam is in the shape of a triangular prism which is 50 m long, as shown.
Both ends of the dam, ABC and DEF , are isosceles triangles with equal sides of 25 metres.
The included angles $\angle BAC$ and $\angle EDF$ are each 150°
Calculate the number of litres of water the dam will hold when full.



7 812 500 L

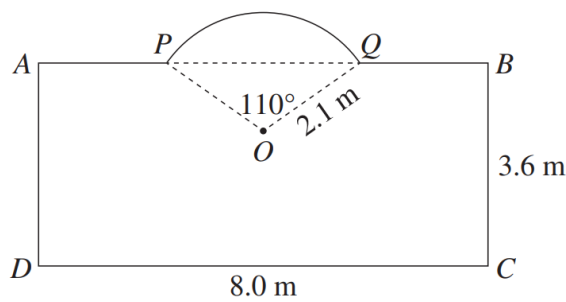
13 2023 HSC Standard 2 Band 5

The diagram shows a shape APQBCD. The shape consists of a rectangle ABCD with an arc PQ on side AB and with side lengths $BC = 3.6$ m and $CD = 8.0$ m.

The arc PQ is an arc of a circle with centre O and radius 2.1 m and $\angle POQ = 110^\circ$

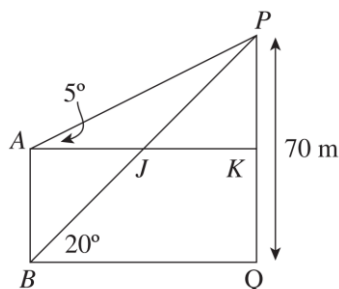
What is the perimeter of the shape APQBCD?

Give your answer correct to one decimal place.



23.8 m

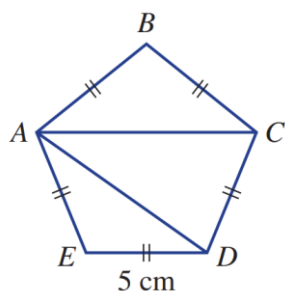
- 14** Two towers AB and PQ stand on level ground. The angles of elevation of the top of the taller tower from the top and bottom of the shorter tower are 5° and 20° respectively. The height of the taller tower is 70 metres.
- Explain why $\angle APJ = 15^\circ$.
 - Show that $AB = \frac{BP \sin 15^\circ}{\sin 95^\circ}$
 - Show that $BP = \frac{70}{\sin 20^\circ}$
 - Hence find the height of the shorter tower, correct to the nearest metre.



53 metres

- 15** Recall that the sum (S) of the interior angles of a polygon with n sides is given by $S = 180(n - 2)$.

Calculate the area of the pentagon. Answer to 1 d.p.

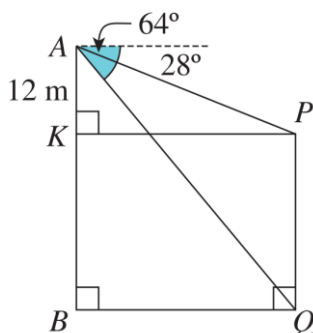


$$\triangle AED = \triangle ABC = 11.89 \text{ cm}^2$$

$$\triangle ADC = 19.24 \text{ cm}^2$$

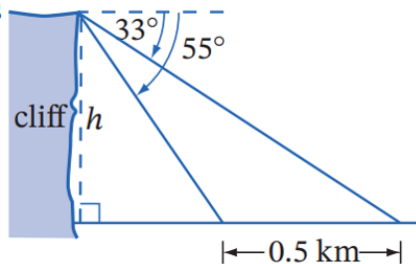
$$43.0 \text{ cm}^2$$

- 16** Two towers AB and PQ stand on level ground. Tower AB is 12 metres taller than tower PQ . From A , the angles of depression of P and Q are 28° and 64° respectively.
- Use $\triangle AKP$ to show that $KP = BQ = 12 \tan 62^\circ$.
 - Use $\triangle ABQ$ to show that $AB = 12 \tan 62^\circ \tan 64^\circ$.
 - Hence find the height of the shorter tower, correct to the nearest metre.
 - Solve the problem again by using $\triangle AKP$ to find AP , and then using the sine rule in $\triangle APQ$.



Shorter tower is 34 metres

- 17** From the top of the cliff, the angles of depression of 2 boats at sea that are 0.5 km apart are 55° and 33° . Show that the height of the cliff $h = \frac{0.5 \sin 33^\circ \sin 55^\circ}{\sin 22^\circ}$



Let the distance of the closer boat to the observer be x .

$$\sin 55^\circ = \frac{h}{x}, x = \frac{h}{\sin 55^\circ}$$

Using sine rule,

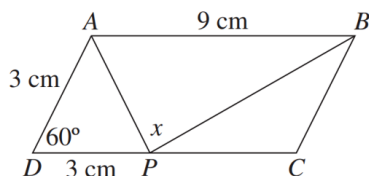
$$\frac{0.5}{\sin 22^\circ} = \frac{x}{\sin 33^\circ}$$

$$\frac{0.5}{\sin 22^\circ} = \frac{\left(\frac{h}{\sin 55^\circ}\right)}{\sin 33^\circ}$$

$$\frac{0.5}{\sin 22^\circ} = \frac{h}{\sin 55^\circ \sin 33^\circ}$$

$$\therefore h = \frac{0.5 \sin 33^\circ \sin 55^\circ}{\sin 22^\circ}$$

- 18** The diagram shows a parallelogram.
- Explain why $\triangle ADP$ is equilateral, and hence find AP .
 - Use the cosine rule in $\triangle BCP$ to find BP in exact form.
 - Let $\angle APB = x$. Show that $\cos x = -\frac{\sqrt{7}}{14}$



a) isosceles triangle, so $\angle ADP = \angle DPA$ and $\angle ADP + \angle DPA = 120^\circ$.

$$\therefore \angle ADP = \angle DPA = 60^\circ$$

Equilateral as all angles 60°

$$\therefore AP = 3 \text{ cm}$$

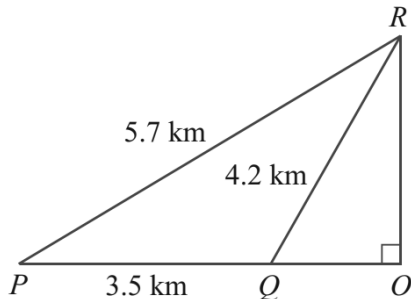
$$\text{b) } 3\sqrt{7} \text{ cm}$$

$$\text{c) } \cos x = \frac{3^2 + (3\sqrt{7})^2 - 9^2}{2(3)(3\sqrt{7})} = \frac{-9}{18\sqrt{7}} = \frac{-9\sqrt{7}}{126} = \frac{-\sqrt{7}}{14}$$

Mixed Trigonometry Problems involving Bearings

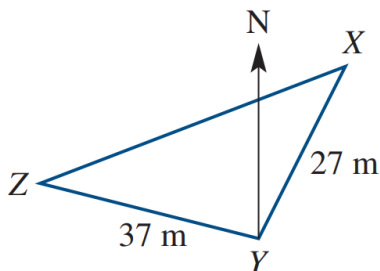
FOUNDATION

- 1 Refer to the diagram to answer the following questions.
- Calculate $\angle PQR$ and hence $\angle RQO$ to the nearest degree.
 - Find the length OR , correct to one decimal place.
 - If OR is North, find the true bearing of Q from R



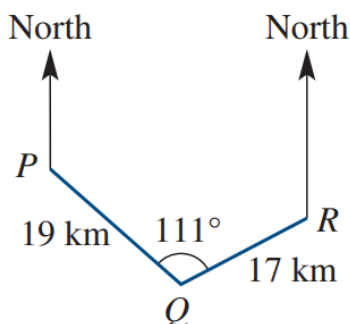
- a) $\angle PQR = 95^\circ, \angle RQO = 85^\circ$
 b) 4.2 km
 c) 185°

- 2 XYZ represents a triangular area of land.
 The bearing of X from Y is 041° and the bearing of Z from Y is 305° .
 The distance XY is 27m and the distance YZ is 37 m.
- What is the size of angle XYZ? Answer correct to the nearest degree.
 - Calculate the area to the nearest square metre.
 - What is the length of the boundary XZ, to the nearest metre?



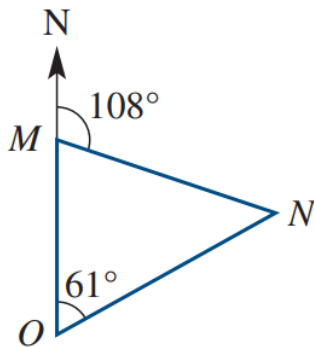
- a) 96°
 b) 497 m^2
 c) 48 m

- 3 The diagram shows information about the locations of towns P, Q and R.
 Amber takes 2 hours and 30 minutes to walk directly from Town P to Town Q.
- What is Amber's walking speed, correct to the nearest km/h?
 - What is the distance from P to R? Answer correct to the nearest kilometre.
 - How long would it take Amber to walk from P to R? Answer to the nearest minute.



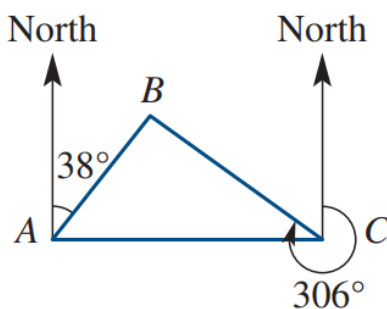
- a) 8 km/h
 b) 30 km
 c) 3 h 45 min

- 4 M is 50km north of O. The bearing of N from M is 108° and from O it is 061° .
 Answer the following questions to the nearest kilometre.
- What is the distance between M and N?
 - What is the distance between N and O?



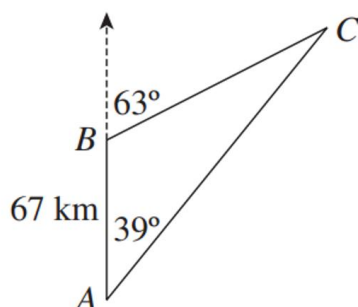
- a) 60 km
 b) 65 km

- 5 A ship is located at B and two radar stations 75 km apart are located at A and C.
 The bearing of the ship from one radar station is 038° and from the other radar station it is 306° .
 C is due east of A.
- How far is the ship from the radar station at A, to the nearest kilometre?
 - How far is the ship from the radar station at C, to the nearest kilometre?



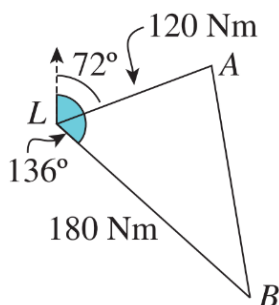
- a) 44 km
 b) 59 km

- 6 A salesman drove from town A to town B, then to town C, then finally directly home to town A.
 Town B is 67 km north of town A, and the bearings of town C from towns A and B are 039°T and 063°T respectively. Find how far the salesman drove, correct to the nearest kilometre.



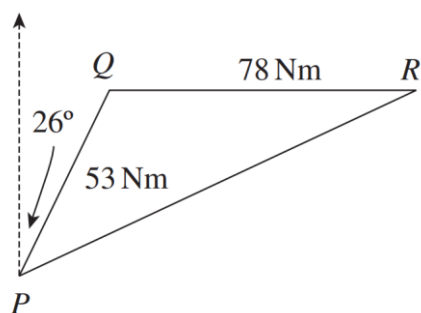
317 km

- 7 In the diagram to the right, ship A is 120 nautical miles from lighthouse L on a bearing of 072°T , while ship B is 180 nautical miles from L on a bearing of 136°T . Calculate the distance between the two ships, correct to the nearest nautical mile.



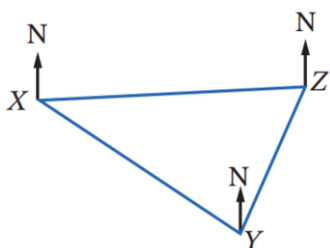
167 nautical miles

- 8 A ship sails 53 nautical miles from P to Q on a bearing of 026°T . It then sails 78 nautical miles due east from Q to R .
- Explain why $\angle PQR = 116^\circ$
 - How far apart are P and R , correct to the nearest nautical mile?

a) draw a north-south line at Q , $\angle PQR = 90 + 26 = 116$ due to alternate angles

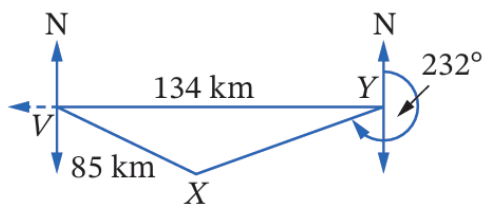
b) 112 nautical miles

- 9 A yacht sails from X to Y on a bearing of 130° for 4.2 km. It then turns and travels to Z on a bearing of 025° for 2.9 km.
- Label the diagram.
 - Explain why $\angle XYZ = 75^\circ$
 - Calculate the distance XZ , correct to 1 d.p.

b) $\angle XYN = 180^\circ - 130^\circ = 50^\circ$ (co-interior angles on parallel lines)

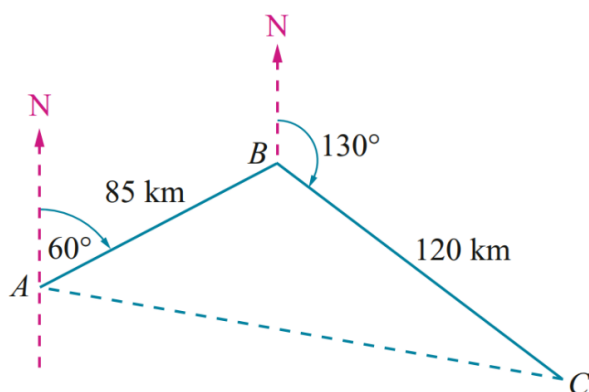
c) 4.4 km

- 10** Yorker (Y) is 134 km due east of Vendor (V).
Xena (X) is on a bearing of 232° from Yorker and is also 85 km from Vendor.
 $\angle VXY$ is an obtuse angle. Find the bearing of Xena from Vendor.



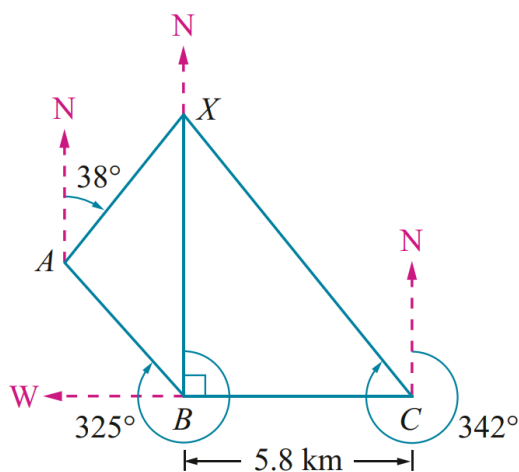
128°

- 11** A ship sails 85 km from A to B on a bearing of 060° T.
It then turns and sails 120 km to C on a bearing of 130° T.
- Find the size of $\angle ABC$.
 - How far is the ship from its starting point?
 - Calculate $\angle BAC$, to the nearest minute.
 - What is the bearing of the ship from its starting point to the nearest minute?



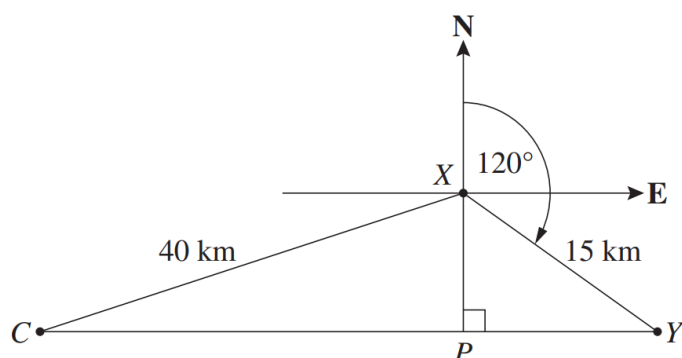
- a) 110°
b) 169 km
c) $41^\circ 49'$
d) $101^\circ 49'$ T

- 12** Consider the diagram.
- Show that the distance BX is 17.9 km.
 - Show that the distance from B to X via A is 22.2 km.



13 2023 HSC Standard 2 Band 5

The diagram shows the location of three places X, Y and C.
 Y is on a bearing of 120° and 15 km from X.
 C is 40 km from X and lies due west of Y.
 P lies on the line joining C and Y and is due south of X.



- a Find the distance from X to P.

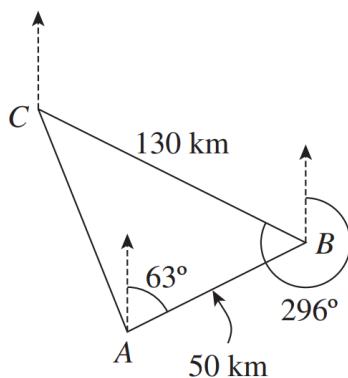
7.5 km

- b What is the bearing of C from X, to the nearest degree?

259°

- 14 A ship sails 50 km from port A to port B on a bearing of 063°T , then sails 130 km from port B to port C on a bearing of 296°T .

- a Show that $\angle ABC = 53^\circ$
 b Find, correct to the nearest km the distance of port A from port C.
 c Find $\angle ACB$, and hence find the bearing of port A from port C, correct to the nearest degree.

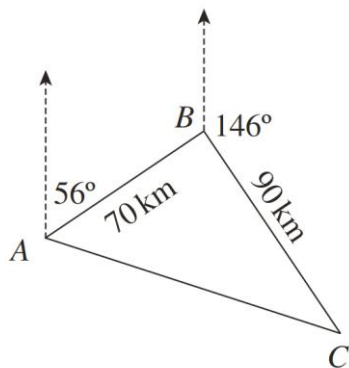


a) extend north line at B, $\angle ABC = 296 - 63 - 180 = 53^\circ$

b) 108 km

c) $\angle ACD \approx 22^\circ$, bearing $\approx 138^\circ$

- 15** A motorist drove 70km from town A to town B on a bearing of 056°T , and then drove 90km from town B to town C on a bearing of 146°T .
- Explain why $\angle ABC = 90^\circ$
 - How far apart are the towns A and C, correct to the nearest kilometre?
 - Find $\angle BAC$, and hence find the bearing of town C from town A, to the nearest degree.



a) angle co-interior to 56° is 124° .

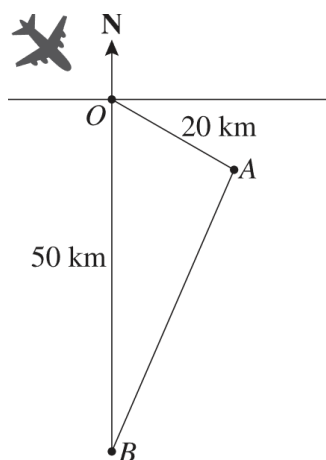
$$\angle ABC = 360 - 124 - 146 = 90^\circ$$

b) 114 km

c) 108°

16 2022 HSC Standard 2 Band 5

The diagram shows an aeroplane that was flying towards an airport at A on a bearing of 135°T . When it was at point O, 20 km away from the airport at A, the flight course was changed. The aeroplane landed at an airport at B directly south of O. The distance from O to B is 50 km.



- Show that the distance between airport A and airport B is 38.5 km, correct to 1 d.p.

38.5 km

- Find $\angle OBA$ to the nearest degree.

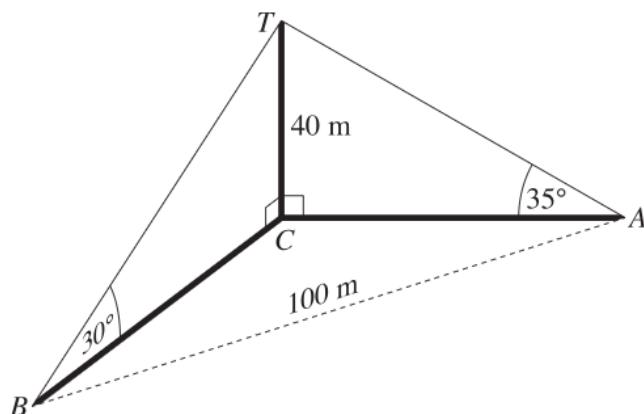
22°

- What is the bearing of the airport at B from the airport at A?

202°

17 2024 HSC Advanced Band 4

A vertical tower TC is 40 metres high. The point A is due east of the base of the tower C . The angle of elevation to the top T of the tower from A is 35° . A second point B is on a different bearing from the tower as shown. The angle of elevation to the top of the tower from B is 30° . The points A and B are 100 metres apart.



- Show that distance AC is 57.13 metres, correct to 2 decimal places.
- Find the bearing of B from C to the nearest degree.

a) In $\triangle TCA$:

$$\tan 35^\circ = \frac{40}{AC}$$

$$AC = \frac{40}{\tan 35^\circ}$$

$$= 57.125\dots$$

$$= 57.13 \text{ m (2 d.p.)}$$

b) In $\triangle TCB$:

$$\tan 30^\circ = \frac{40}{BC}$$

$$BC = \frac{40}{\tan 30^\circ}$$

$$= 69.28 \text{ m}$$

Find $\angle BCA$ using cosine rule:

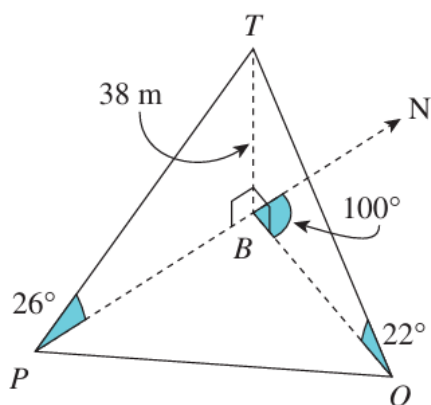
$$\cos \angle BCA = \frac{57.13^2 + 69.28^2 - 100^2}{2(57.13)(69.28)}$$

$$= -0.2446\dots$$

$$\angle BCA = 104.2^\circ$$

$$\therefore \text{Bearing of } B \text{ from } C = 90 + 104 = 194^\circ$$

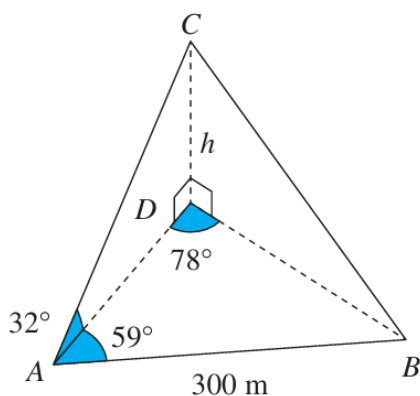
- 18** From two points P and Q on level ground, the angles of elevation of the top T of a 38 m tower are 26° and 22° respectively.
Point P is due south of the tower, and the bearing of Q from the tower is 100° .
a Show that $PB = 38 \tan 64^\circ$, and find a similar expression for QB .
b Hence determine, correct to the nearest metre, the distance between P and Q .



- a) $38 \tan 68^\circ$
b) 111 m

- 19** Two observers at A and B on horizontal ground are 300 m apart.
From A , the angle of elevation of the top C of a tall building DC is 32° .
It is also known that $\angle DAB = 59^\circ$ and $\angle ADB = 78^\circ$.

- a** Show that $AD = \frac{300 \sin 43^\circ}{\sin 78^\circ}$
b Hence find the height of the building, correct to the nearest metre.

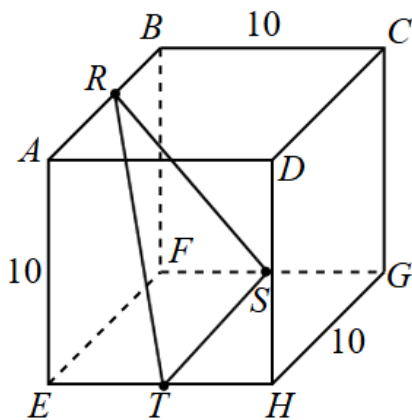


131 m

20 HSC Sample Question Band 5

A cube ABCDEFGH is pictured below. R, S, and T are the midpoints of AB, FG and EH as shown. Side length = 10.

Calculate the size of the angle TRS, giving your answer to one decimal place.



$$ST = 10$$

Find RT using $\triangle RQT$ where Q is the midpoint of EF:

$$RQ = 10, QE = ET = 5$$

$$QT = \sqrt{5^2 + 5^2} = \sqrt{50}$$

$$RT = \sqrt{10^2 + (\sqrt{50})^2} = \sqrt{150}$$

$$\text{By symmetry, } RS = \sqrt{150}$$

Using cosine rule in $\triangle RST$:

$$\cos \angle TRS = \frac{(\sqrt{150})^2 + (\sqrt{150})^2 - 10^2}{2 \times \sqrt{150} \times \sqrt{150}} = \frac{2}{3}$$

$$\therefore \angle TRS = \cos^{-1}\left(\frac{2}{3}\right) = 41.18 \dots^\circ = 41.2^\circ$$

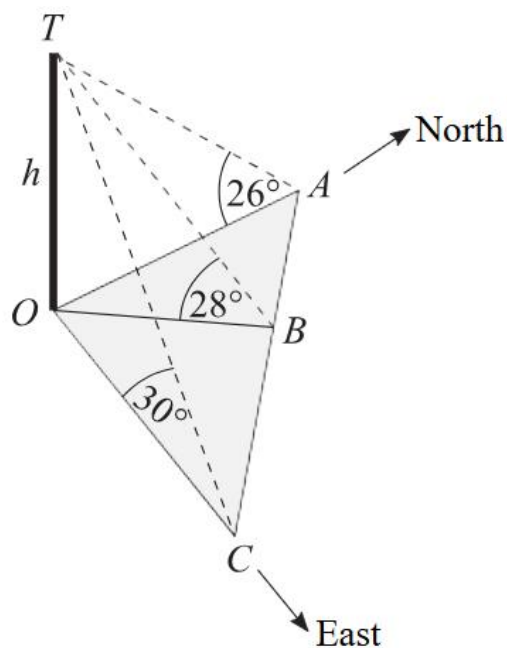
21 HSC Sample Question Band 6

A tower is built on flat ground.

Three tourists, A, B and C are observing the tower from ground level. A is due north of the tower, C is due east and B is on the line of sight from A to C and between them.

The angles of elevation of the top of the tower from A, B and C are 26° , 28° and 30° , respectively.

What is the bearing of B from the tower?



Let h = height of tower

$$\text{In } \triangle OAT: \tan 26^\circ = \frac{h}{d_A}$$

$$d_A = \frac{h}{\tan 26^\circ}$$

$$\text{Similarly, } d_B = \frac{h}{\tan 28^\circ}, d_C = \frac{h}{\tan 30^\circ}$$

In $\triangle OAC$ (right-angled, A due north, C due east):

$$\tan \angle OAC = \frac{d_C}{d_A} = \frac{\frac{h}{\tan 30^\circ}}{\frac{h}{\tan 26^\circ}} = \frac{\tan 26^\circ}{\tan 30^\circ} = 0.8447 \dots$$

$$\angle OAC = 40.19^\circ$$

Using sine rule in $\triangle OAB$:

$$\sin \frac{\angle ABO}{d_A} = \sin \frac{\angle OAC}{d_B}$$

$$\sin \angle ABO = \sin 40.2^\circ \times \frac{\tan 28^\circ}{\tan 26^\circ} = 0.7035 \dots$$

$$\angle ABO = 44.71^\circ \text{ or } 135.29^\circ$$

$$\text{Since } \angle OCA = \tan^{-1} \left(\frac{\tan 30^\circ}{\tan 26^\circ} \right) = 49.8^\circ$$

$$\angle OBC = 44.71^\circ \text{ (otherwise angle sum } \triangle OBC > 180^\circ)$$

$$\angle AOB = 180 - (40.19 + 135.29) = 4.52^\circ$$

$$\therefore \text{Bearing of B from tower} = 005^\circ$$